

Basic Tissues

Epithelium and CT



Ordinary Connective Tissue Cells

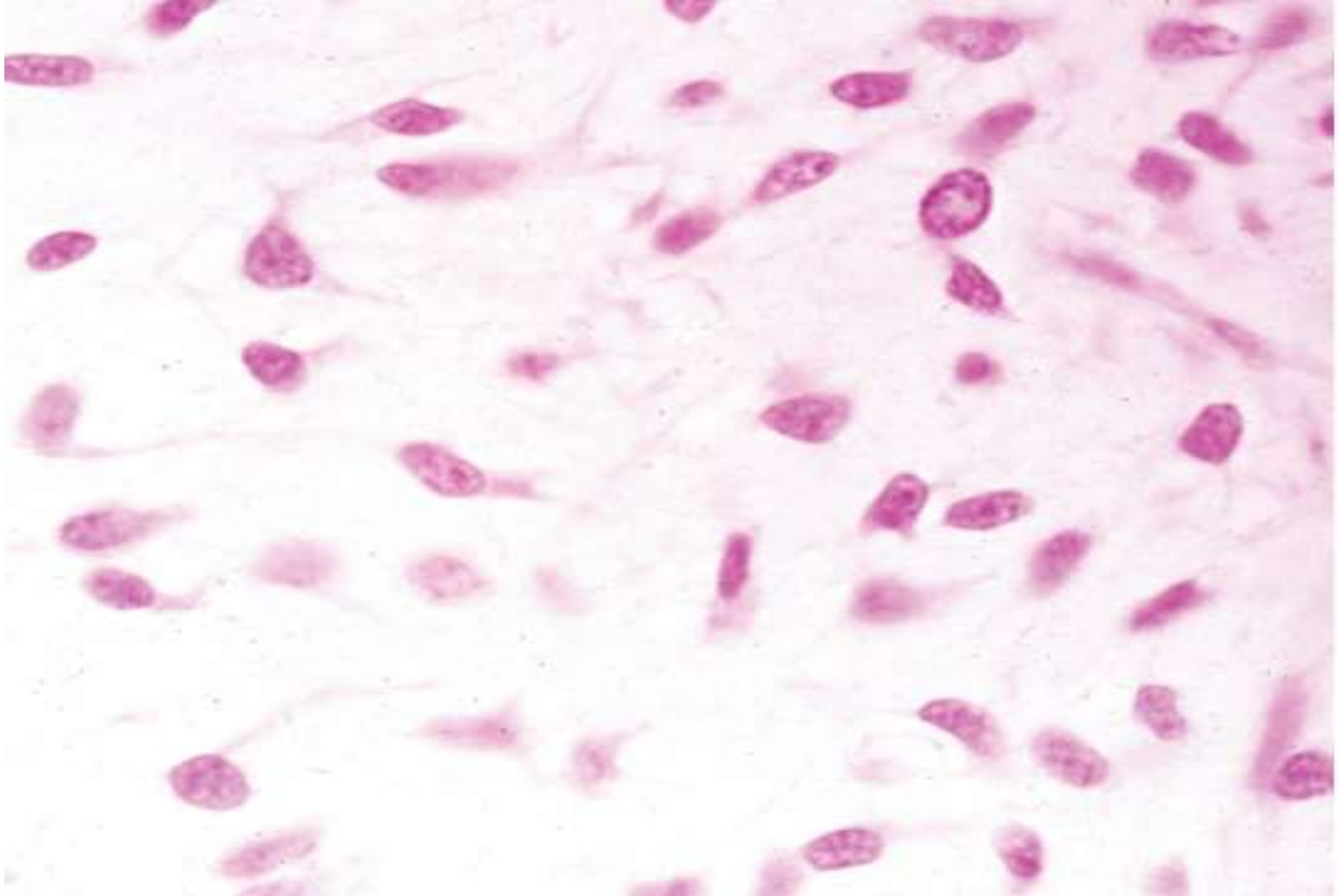
Fixed (Resident) cells

- Fibroblasts and fibrocytes
- Reticular cells
- Myofibroblasts (observed during wound healing)
- Adipocytes/Fat cells (white or yellow and brown)
- Mast cells
- Fixed macrophages (histiocytes), mesenchymal cells

Transient or Wandering cells

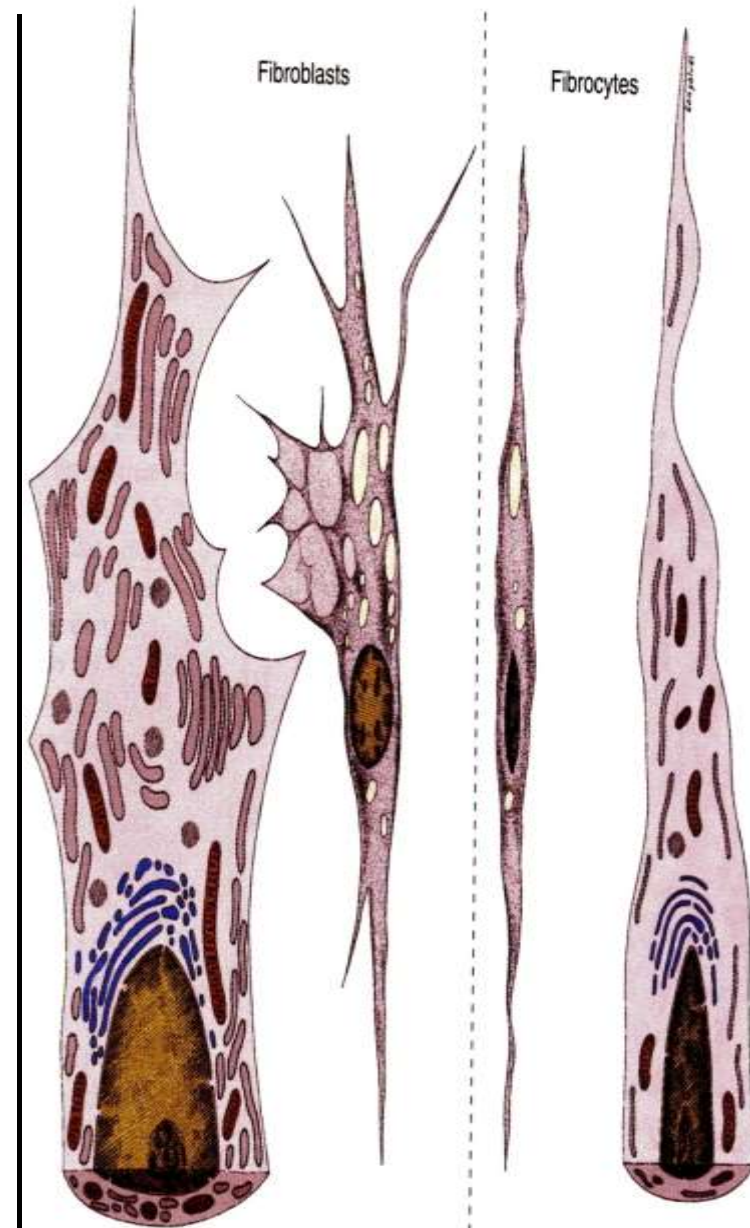
- Plasma cells
- Neutrophils
- Eosinophils
- Basophils
- Lymphocytes, monocytes & Transient macrophages

Mesenchyme: consists of mesenchymal cells surrounded by an extracellular matrix which they produced and which in turn consists largely of a simple loose ground substance rich in hyaluronon (hyaluronic acid). This section is stained with Masson trichrome which stains collagen fibers green and the lack of collagen in mesenchyme is apparent.

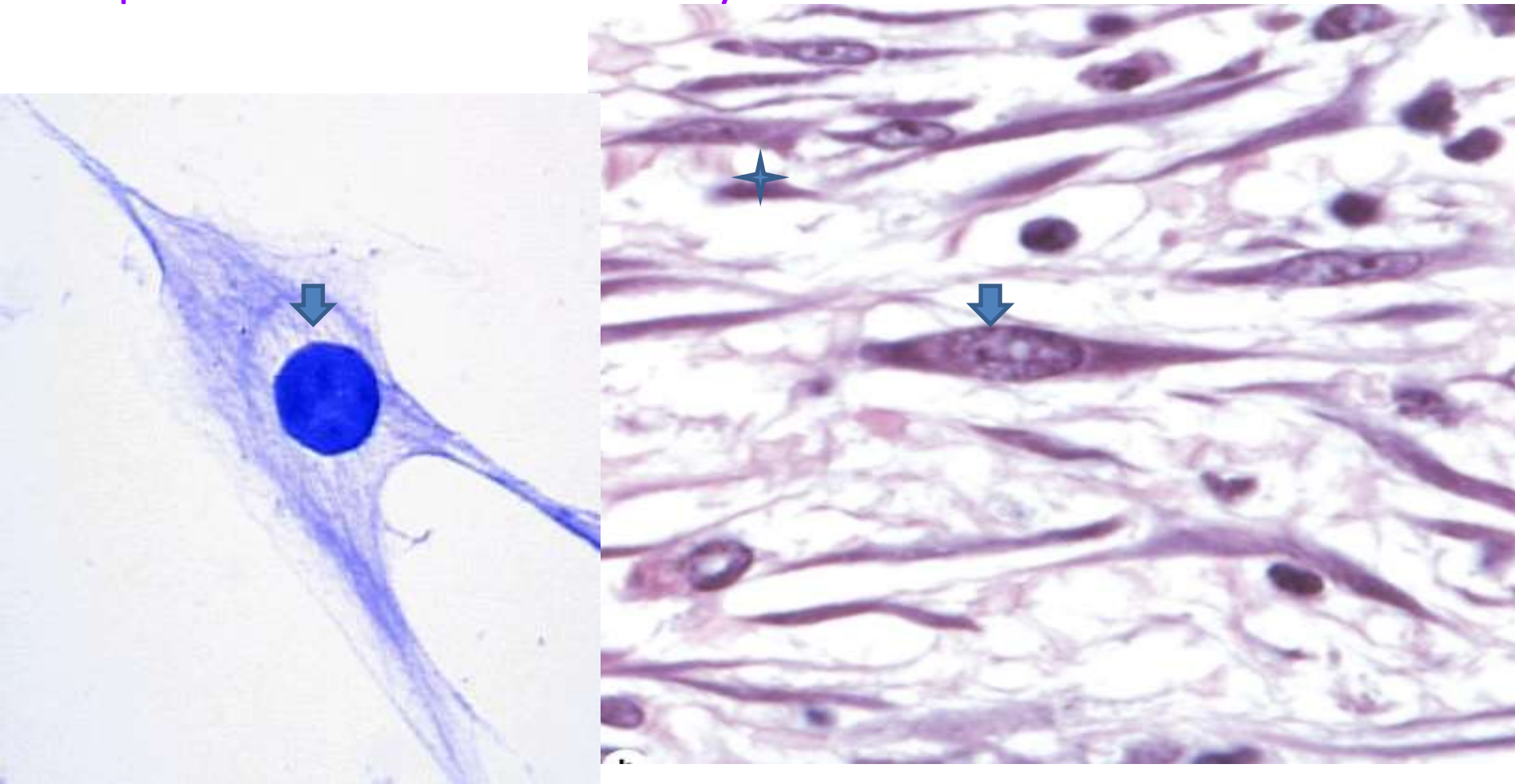


FIBROBLASTS

- Fibro = fiber, blast = forming/making
- Elongated fusiform cells with pale staining cytoplasm and large granular, ovoid nucleus with prominent nucleoli reflecting active protein synthesis.
- produce extracellular matrix (fibers and ground substance) in ordinary CTs.
- derived from undifferentiated mesenchymal cells.
- Present nearby collagen fibers.
- **Fibrocytes** are inactive/quiescent forms of fibroblasts characterized by flattened condensed nucleus.

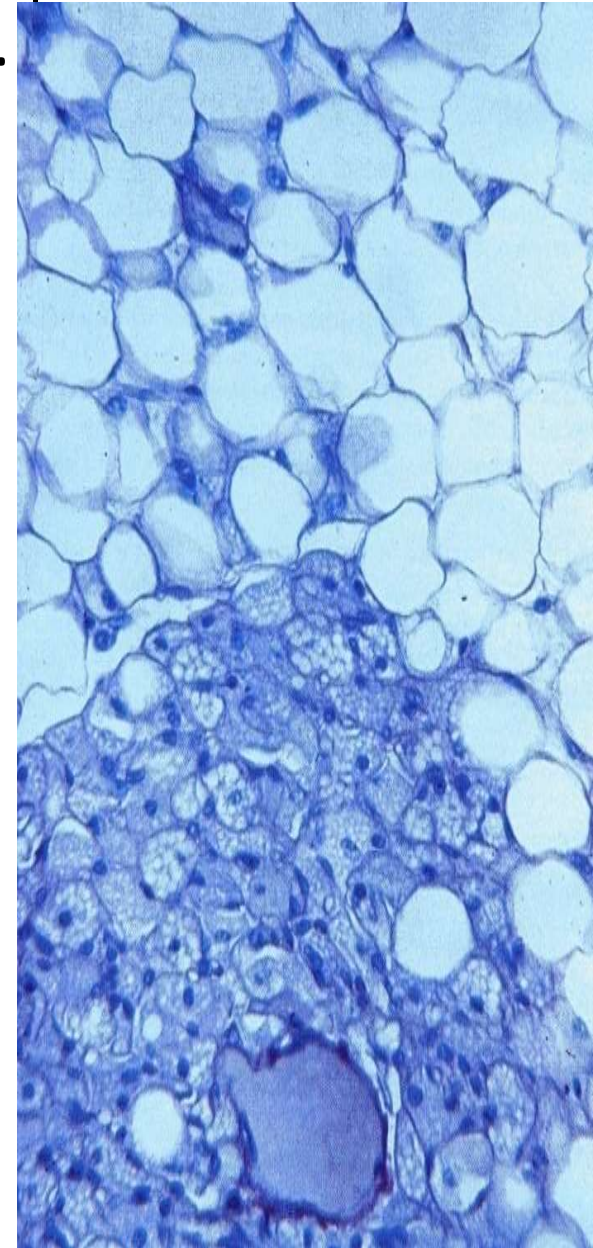


Both active (arrow) and quiescent fibroblasts (fibrocytes★) may sometimes be distinguished, as in this section of dermis. Active fibroblasts are large cells with large, euchromatic nuclei and basophilic cytoplasm, whereas inactive fibroblasts or fibrocytes are smaller with less prominent, heterochromatic nuclei. The very basophilic round cells are leukocytes.



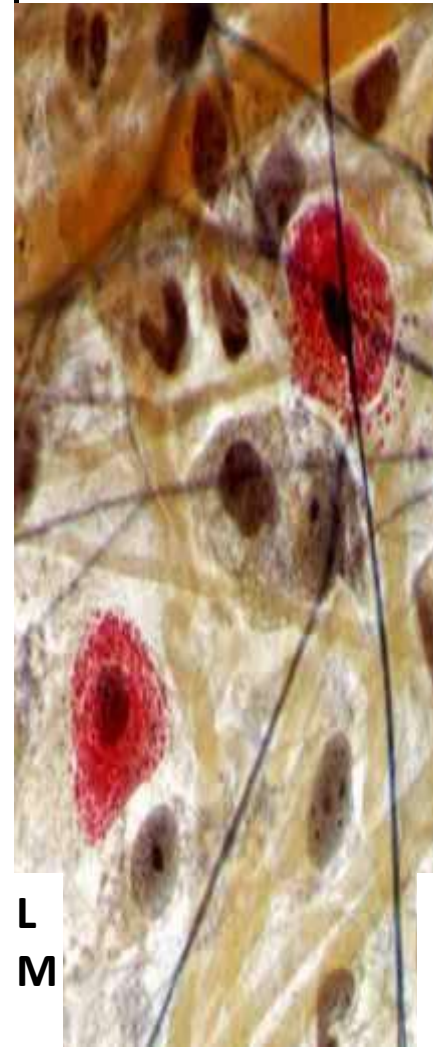
ADIPOCYTES (L. adeps, fat, cyte= cell)

- synthesize, store & release fat and leptin.
- appear individually or in small clusters in some fibrous CTs.
- If they dominate an area, we call that area adipose tissue.
- Contain huge droplet(s) of lipids for storage
- derived from mesenchymal cells.
- Each cell rests on an **external lamina** (homologous to basal lamina of epithelium) which consists of reticular fibers, capillaries & sympathetic nerves
- they are of two types: **white & brown adipocytes.**

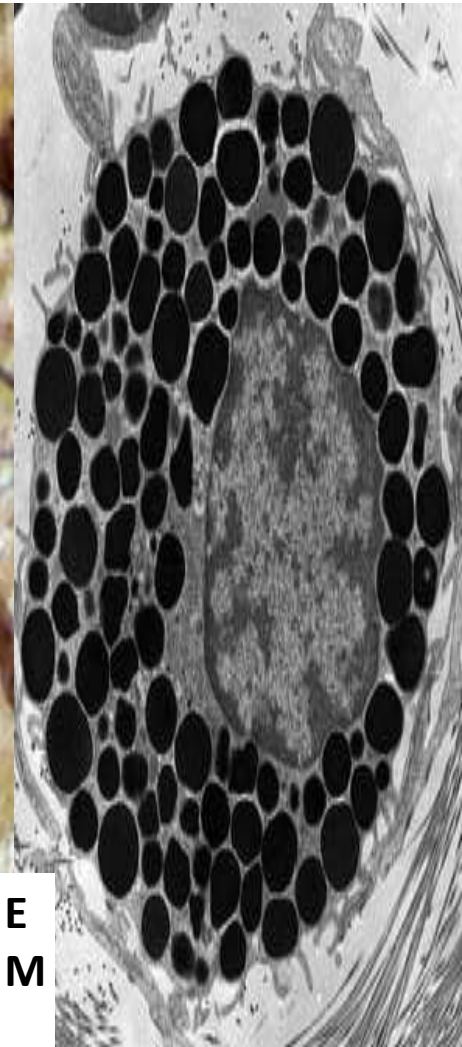


MAST CELLS/MASTOCYTES (HISTAMINOCYTES)

- are large oval or round cells about 20 μm in diameter, whose cytoplasm is filled with basophilic secretory granules.
- The rather small, spherical nucleus is centrally situated and may be obscured by the coarse cytoplasmic granules.
- They are derived from precursors in bone marrow.
- found in loose CT adjacent to blood vessels and body surfaces (dermis of skin and lamina propria of digestive and respiratory tracts).
- Their cytoplasm have large basophilic metachromatic granules containing **heparin, histamine** and many other mediators.



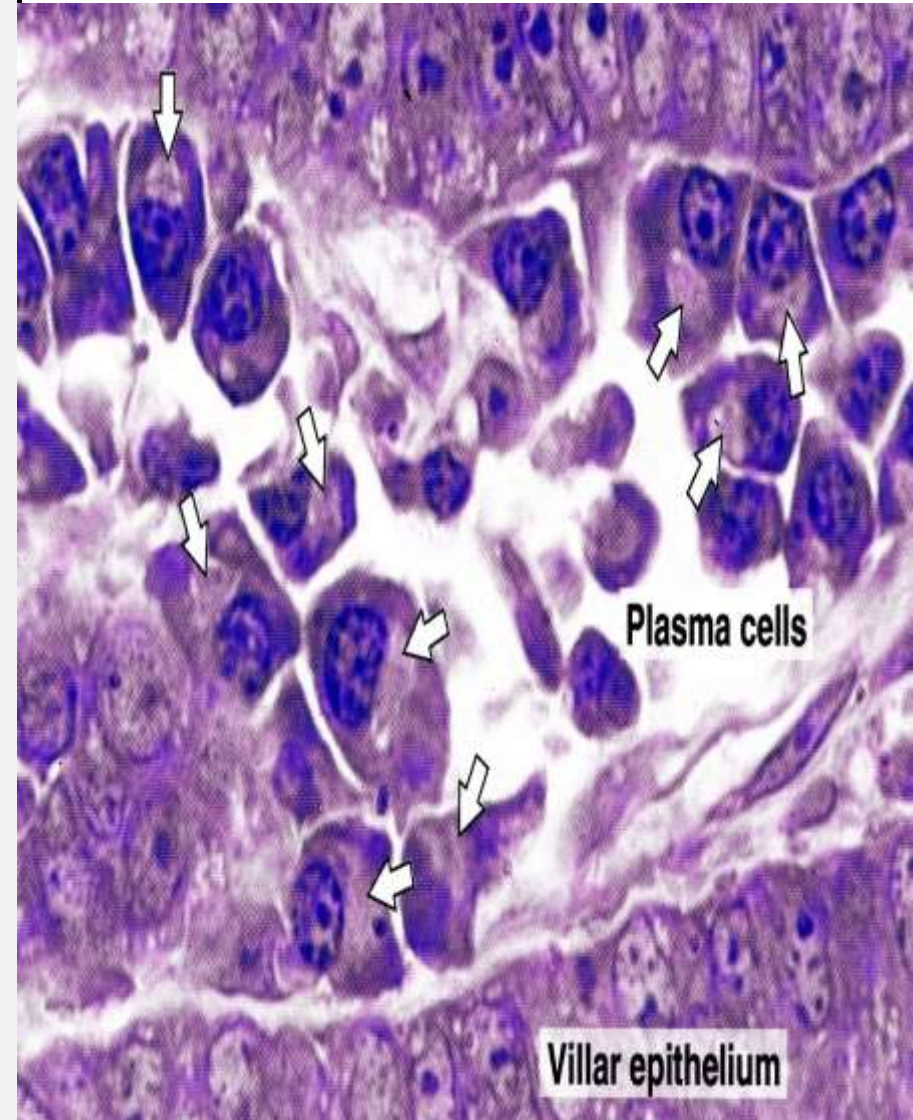
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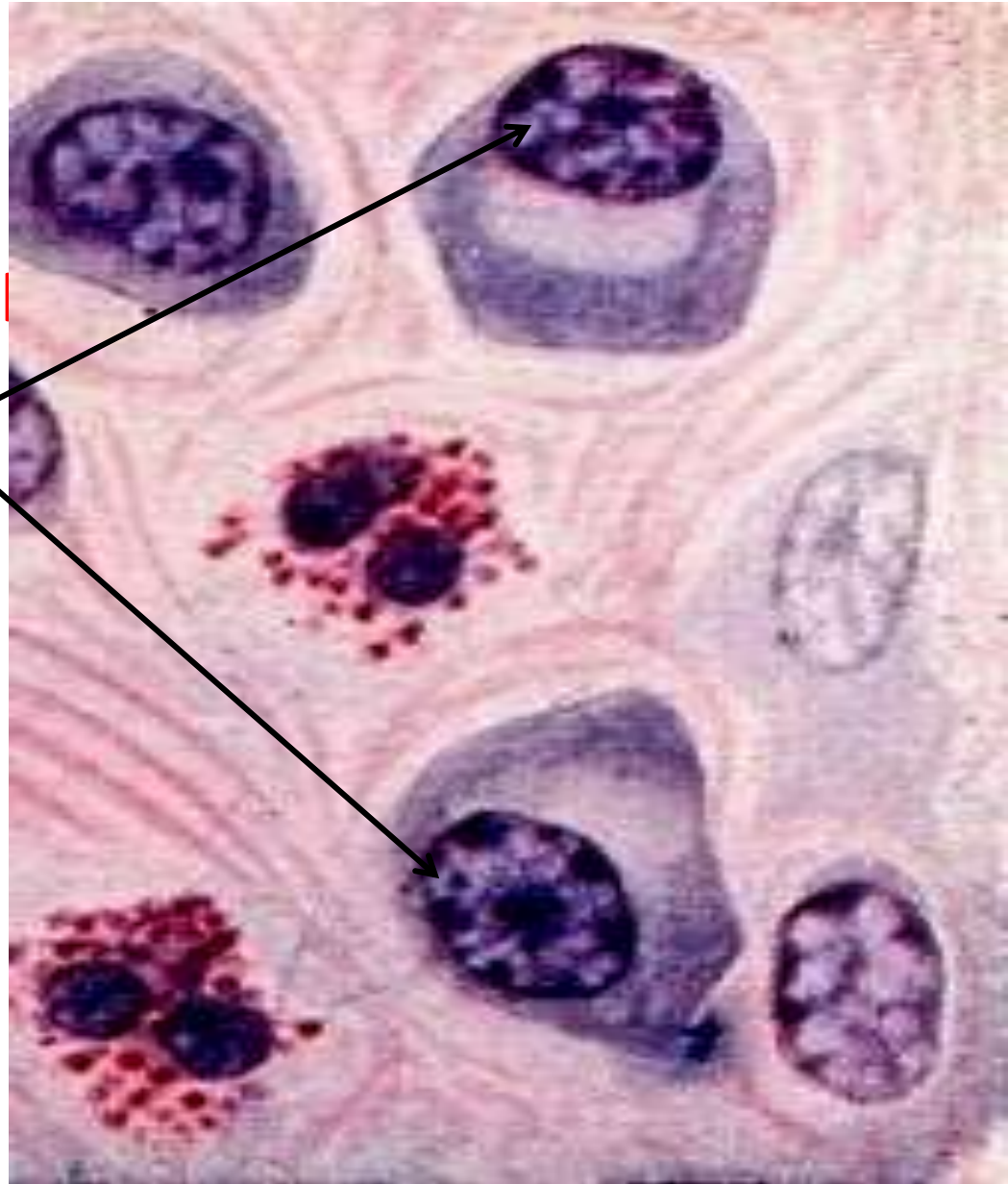
PLASMA CELLS

- Large ovoid cells with eccentrically placed nucleus.
- They are derived from activated B-lymphocytes.
- reveal intensely basophilic cytoplasm due to abundant RER and ribosomes.
- nucleus has the characteristic “clock-face” or “radial spoke” appearance.
- present in areas of chronic inflammation and foreign bodies, where they manufacture and secrete antibodies.

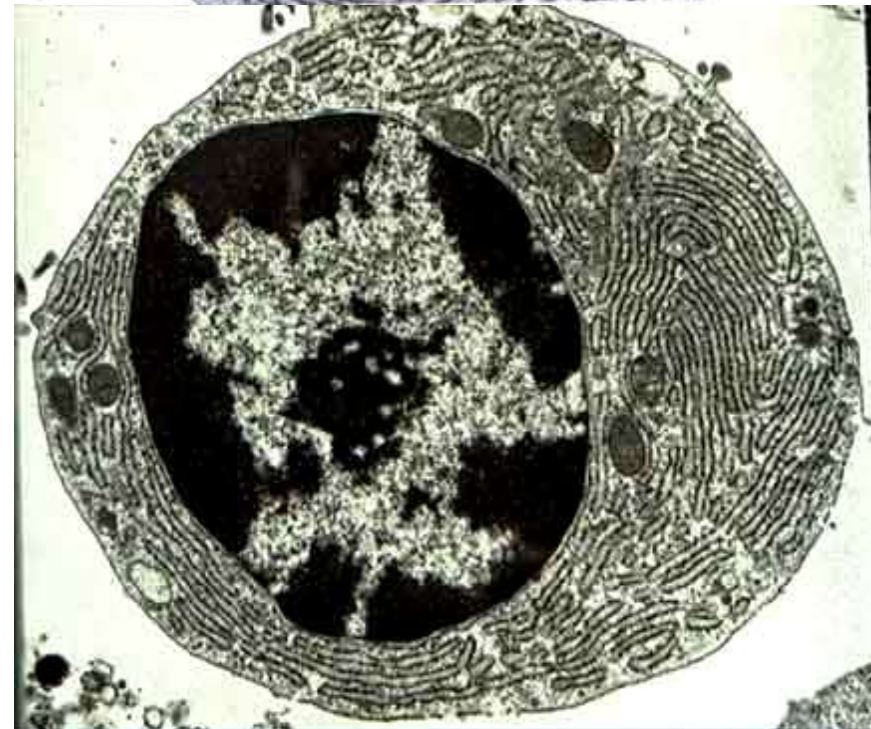
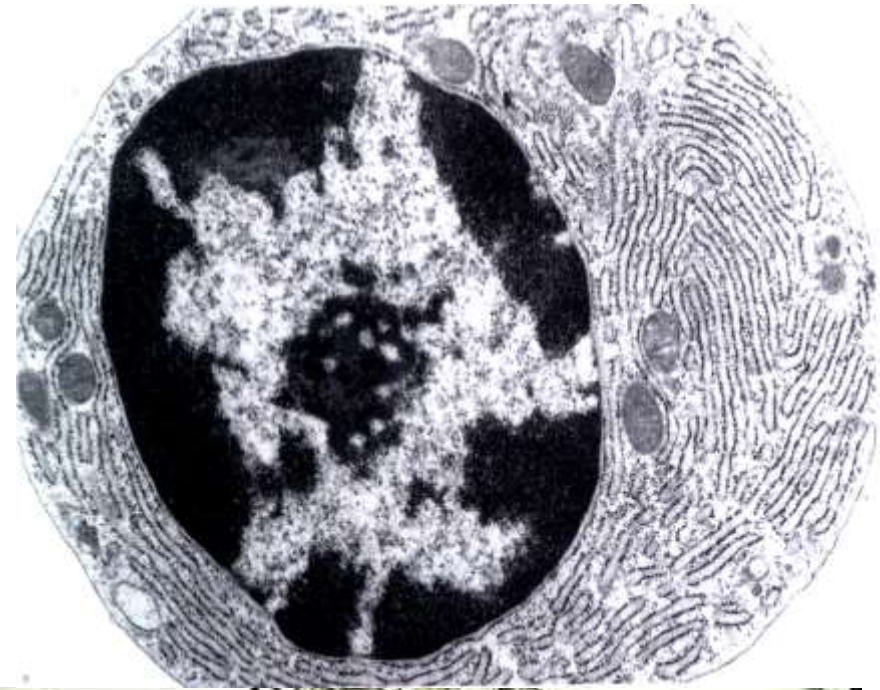
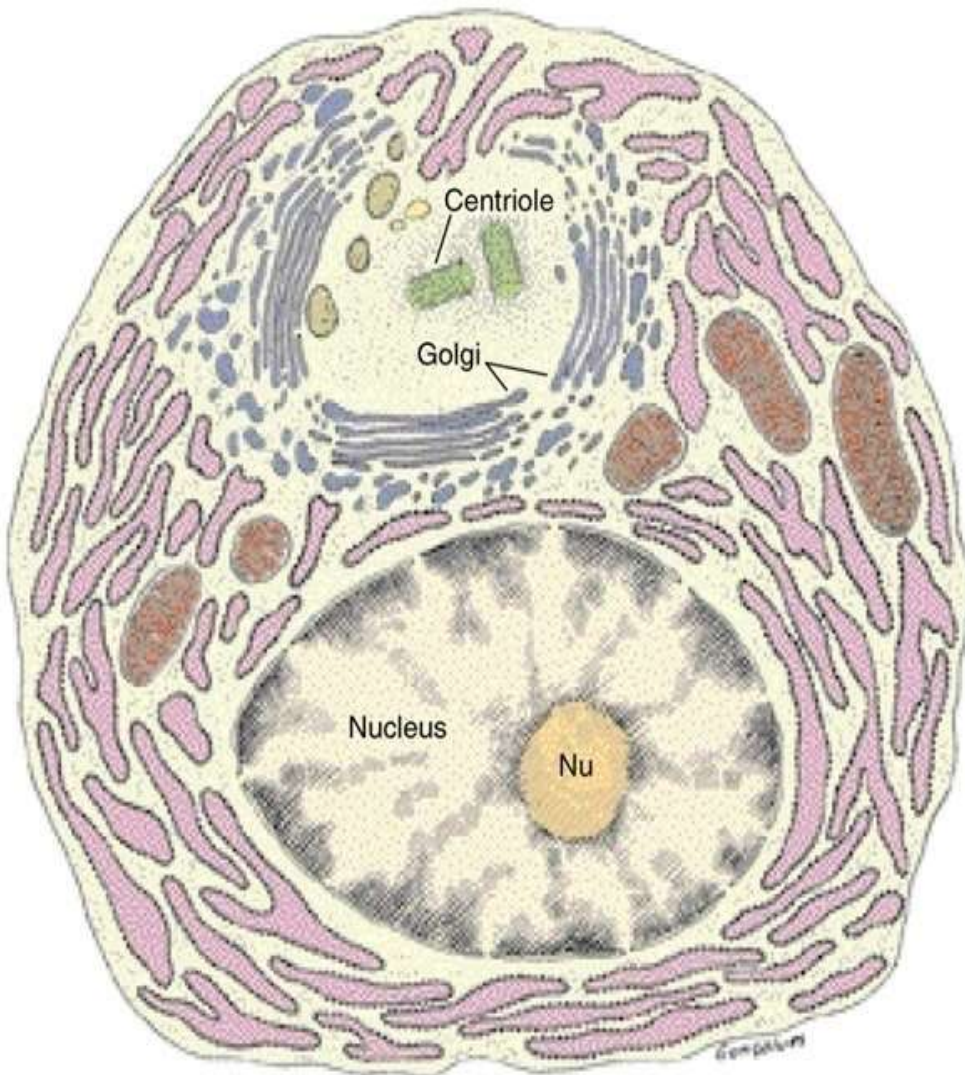


Features of Plasma cells

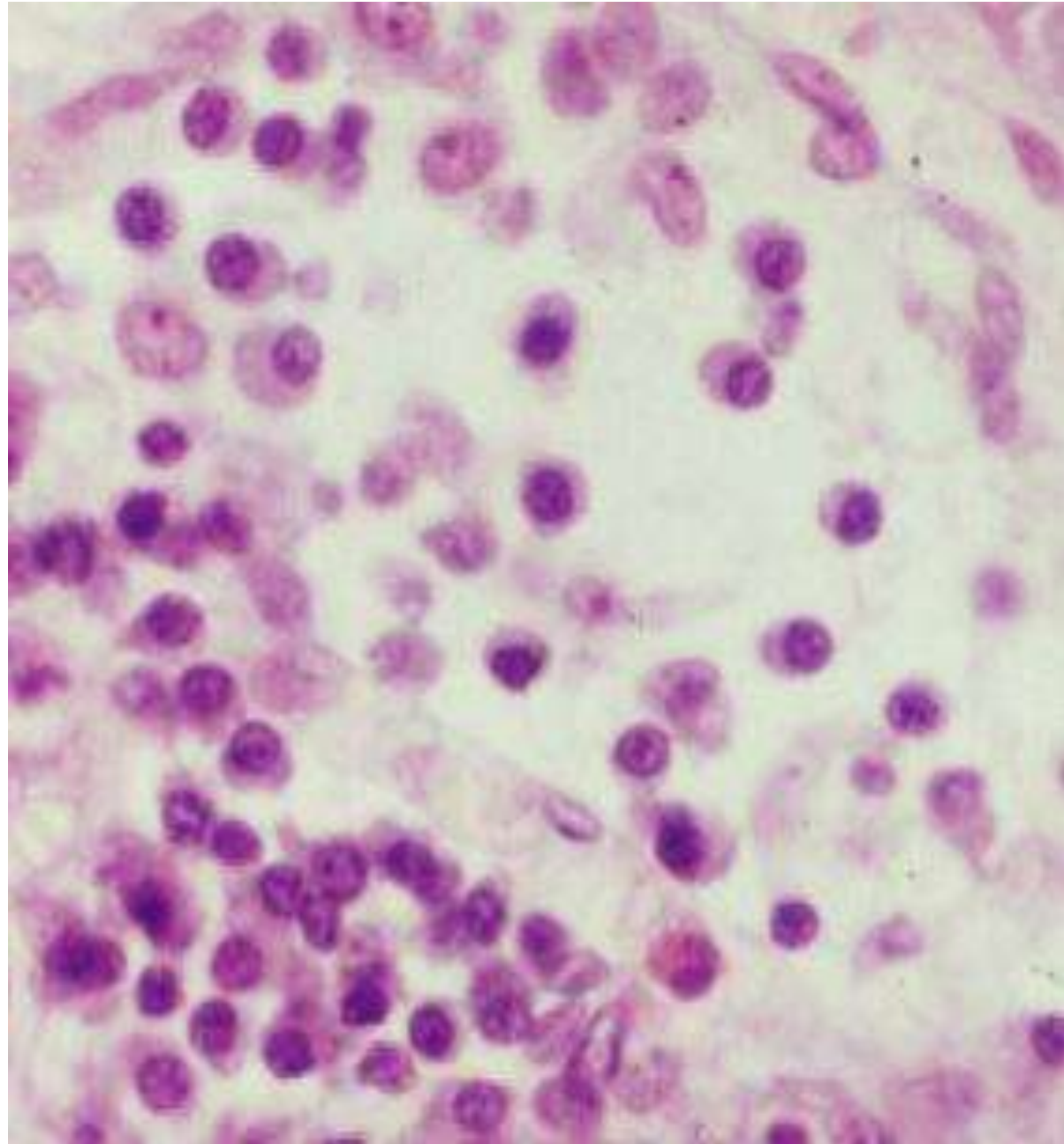
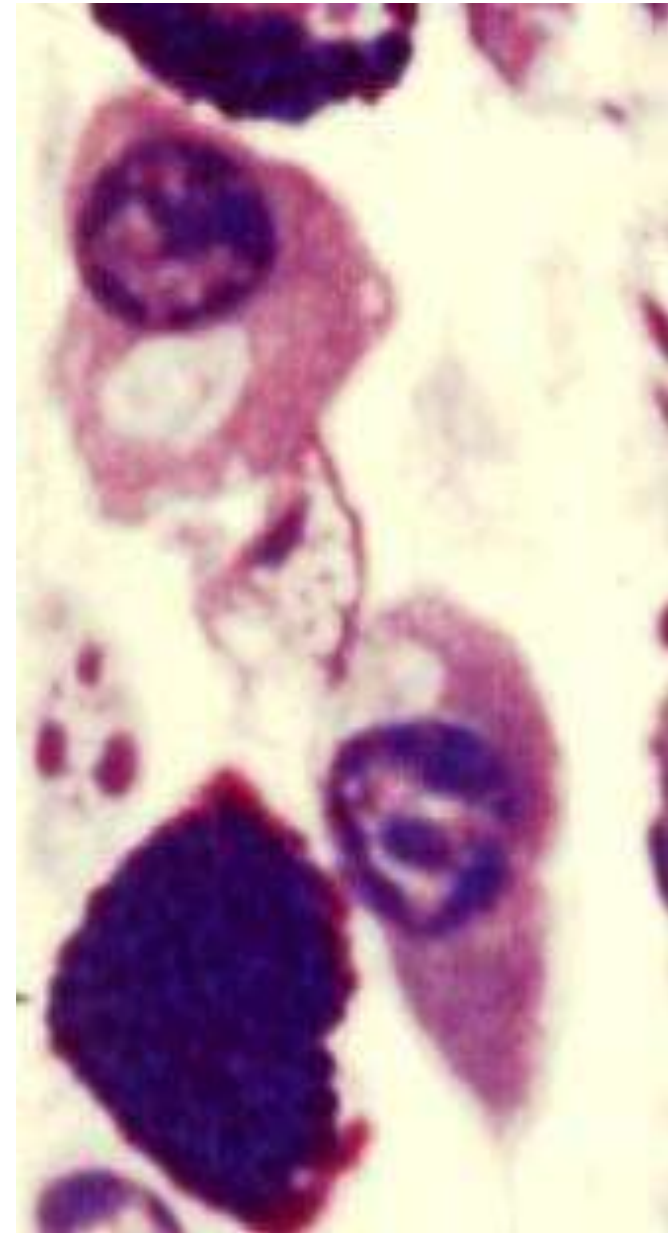
- are typical protein secreting cells
- basophilic cytoplasm
- eccentrically placed **radial spoke** or **clock face** or **cartwheel nucleus** partially surrounded by a single large Golgi apparatus
- effectors cells of immune system derived from B cells
- produce immunoglobulin antibodies



Plasma cell: typical protein secreting cell

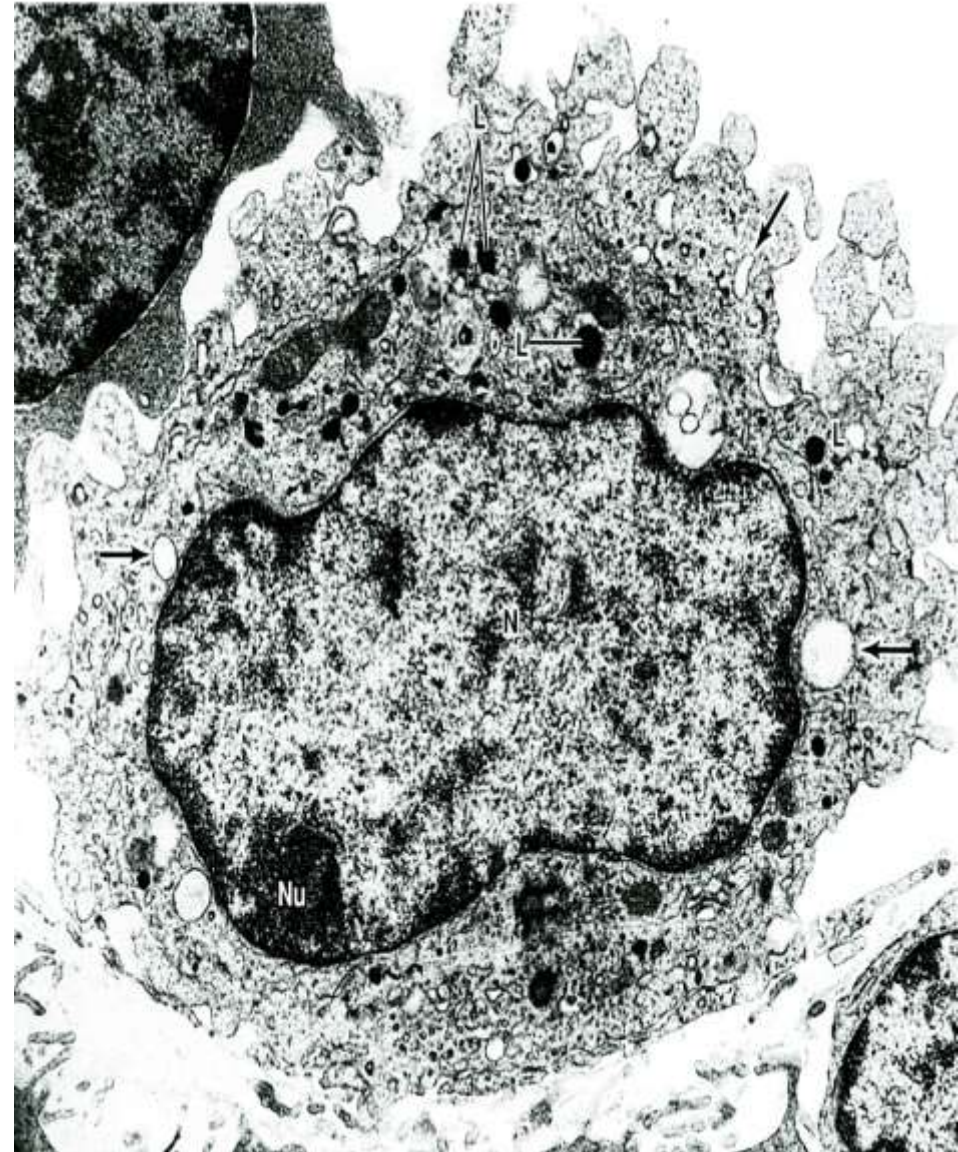


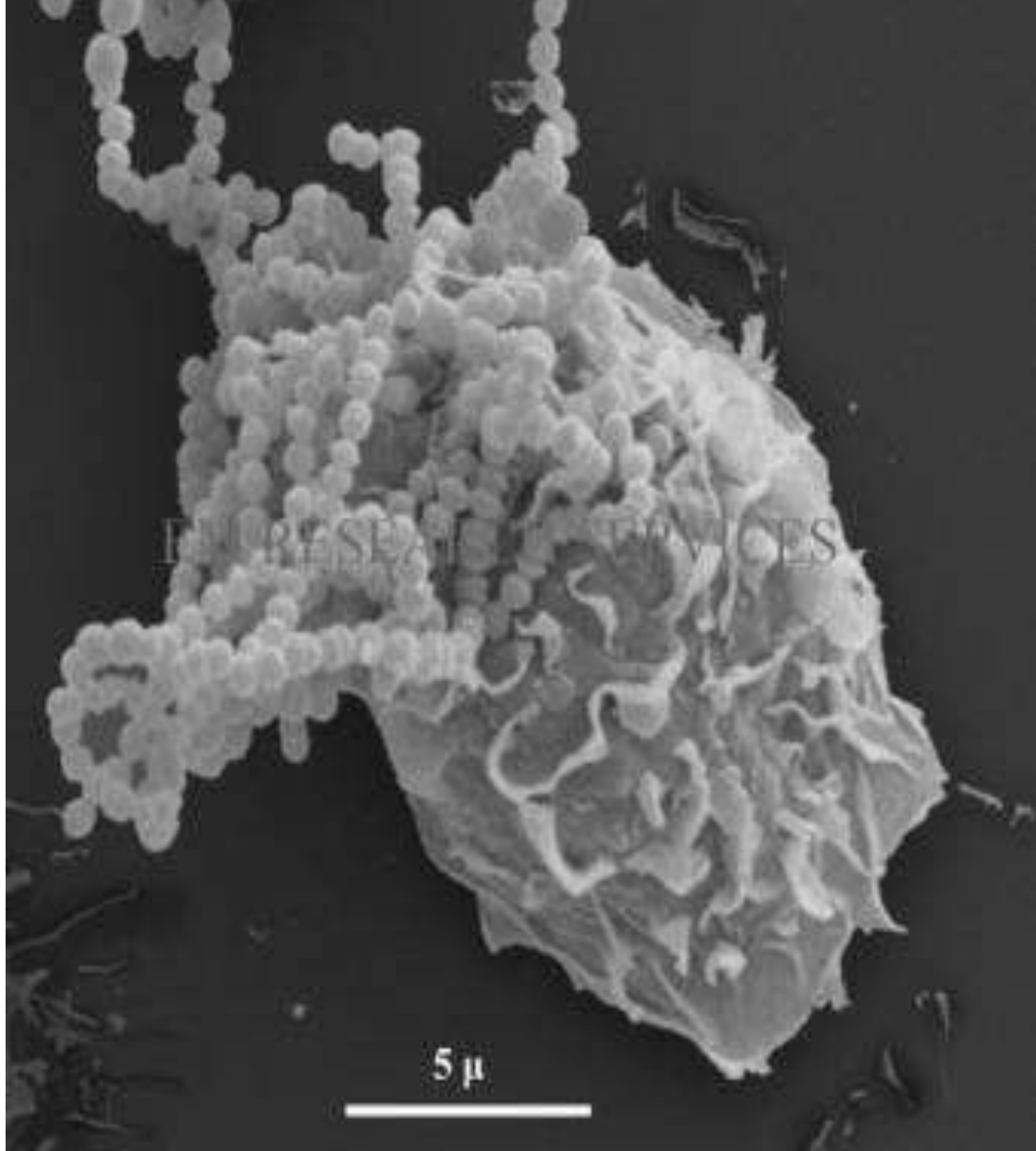
Plasma cells, lymphocytes, mast cells,



MACROPHAGES/HISTIOCYTES/CLASMATOCYTES

- They may be fixed or transient.
- They are derived from monocytes (transient) or from hematopoietic stem cells (fixed macrophages).
- A typical macrophage measures between 10 and 30 μ m in diameter and has an oval or kidney-shaped nucleus located eccentrically.
- Highly motile, irregular in shape with pseudopodia.
- Have dark nucleus and basophilic cytoplasm containing numerous lysosomes, vacuoles and residual bodies.

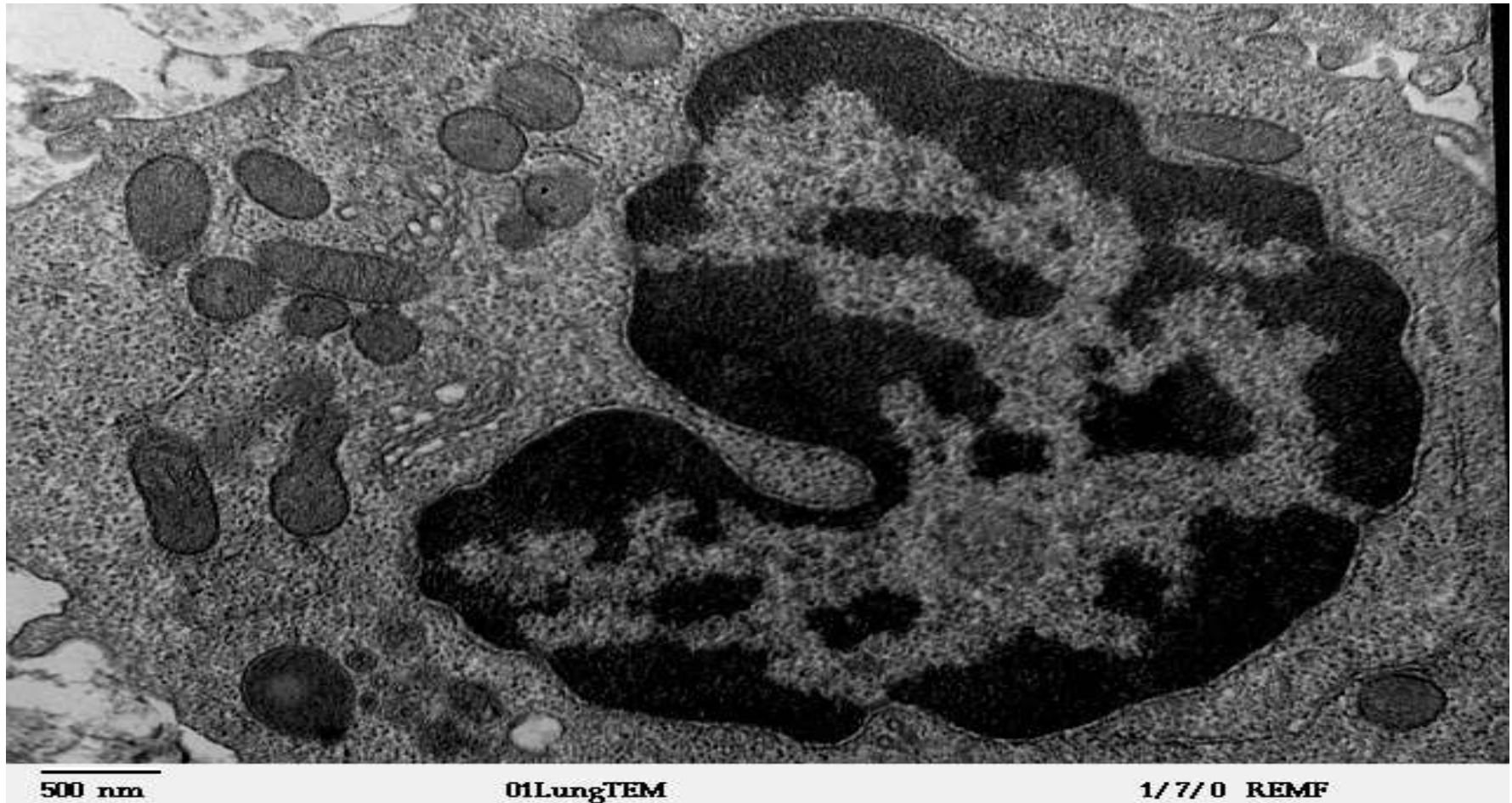




MACROPHAGE BY EM

- They are active phagocytes removing cell debris and protecting the body against foreign invaders.
- As shown here, a microphage is undergoing phagocytosis of streptococci bacteria

MACROPHAGE BY TEM



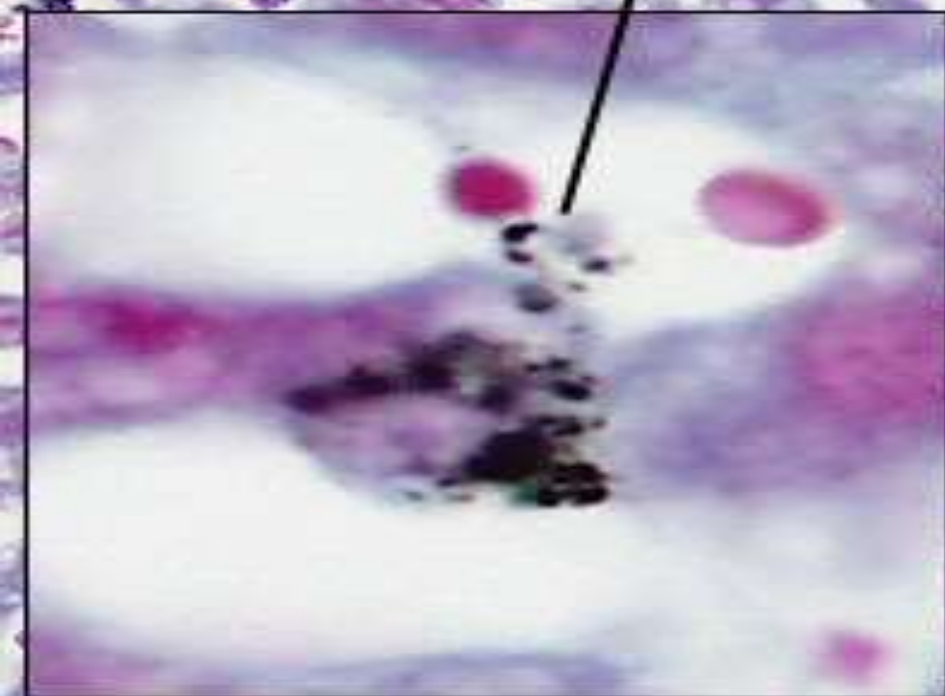
- A macrophages characterized by an irregular surface with pleats, protrusions, and indentations, a morphologic expression of their active pinocytotic and phagocytic activities.
- Has generally well-developed Golgi apparatus, many lysosomes, and rough ER.

Distribution & Main Functions of the Cells of the Mononuclear Phagocytic System

Cell Type	Location	Main Function
Monocytes	Blood	Precursor of macrophages
Tissue macrophages (Histiocytes)	All Connective tissues, lymphoid organs, bone marrow	Production of cytokines, chemotactic factors, and several other molecules that participate in inflammation (defence), antigen processing and presentation
<i>Kupffer</i> cell	Liver	Same as macrophages
Microglial cell	Nerve tissue of the CNS	Same as macrophages
Dust cells	Lung tissue	Same as macrophages
Langerhans cells	Epidermis of Skin	Antigen processing and presentation
Microfold cells	Peyer's patch of GIT	Antigen processing and presentation
Dendritic cells	Lymph nodes	Antigen processing and presentation
Osteoclasts	Bone (formed by fusion of several macrophages)	Digestion of bone matrix to produce bone marrow cavities
Multinuclear giant cell	Connective tissue (fusion of several macrophages)	Segregation and digestion of foreign bodies

Liver ink injected, trichrome

macrophages



Classification of CT

I. Embryonic CTs:

1. Mesenchyme

2. Muroid (Mucous CT)

B. Specialized CTs/ CTs with special properties:

1. Blood tissue

2. Bone tissue

3. Cartilage

II. Mature CT proper:

A. Ordinary/Fibrous CTs

1. Loose CT

- a. Loose areolar
- b. Adipose
- c. Reticular

2. Dense CT

- a. Dense regular
 - 1. Collagenous
 - 2. Elastic
- b. Dense irregular
 - 1. Collagenous
 - 2. Fibroelastic

Types of Fibrous Connective Tissues

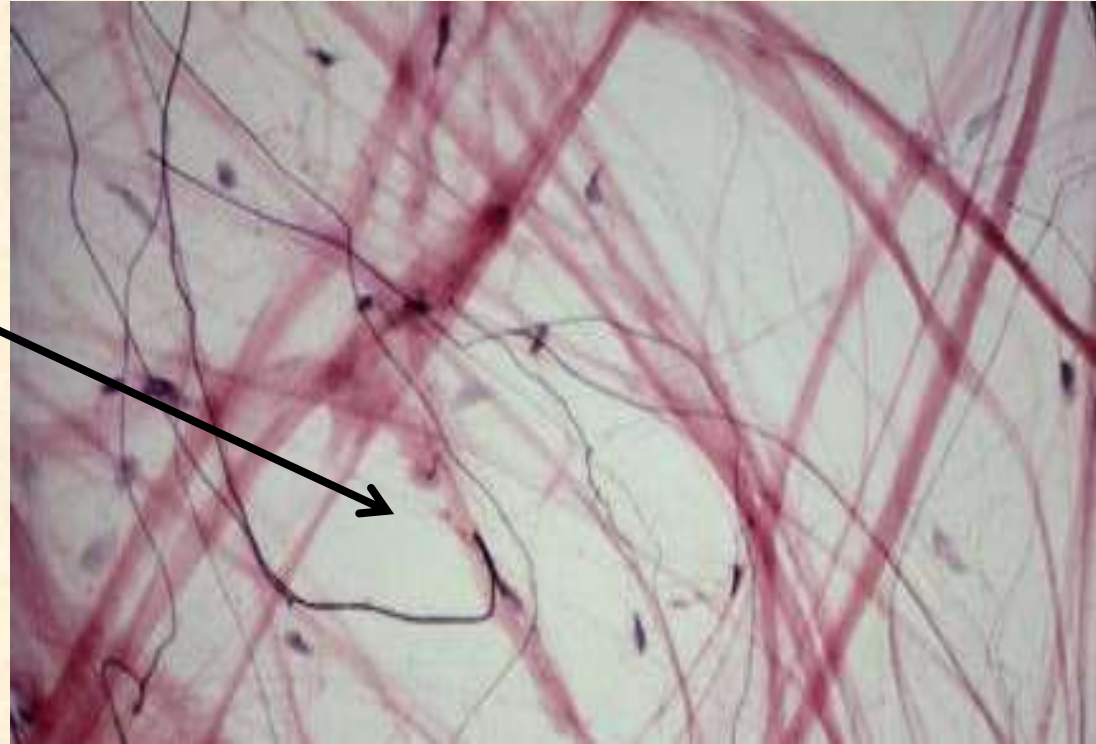
- Two types based on the relative abundance of fibers:

1. Loose Connective Tissue

- Lots of ground substance and cells. Fewer fibers.
- Leaves abundant empty spaces in tissue sections.

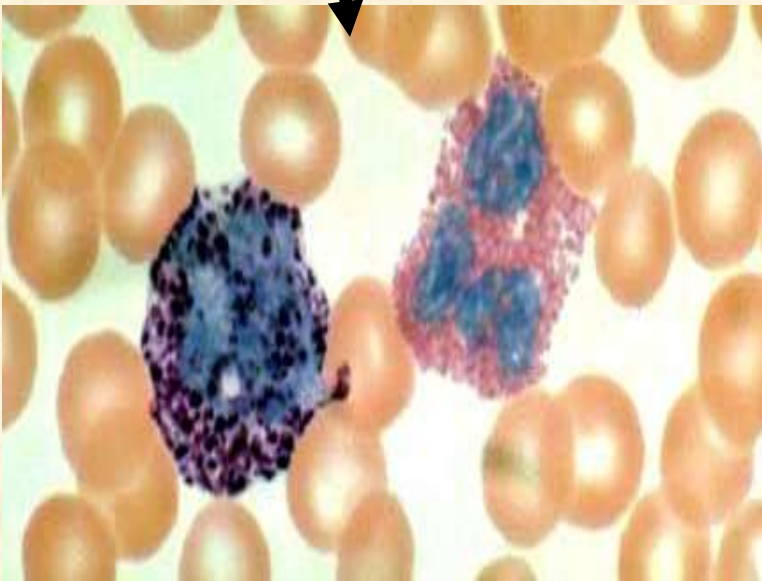
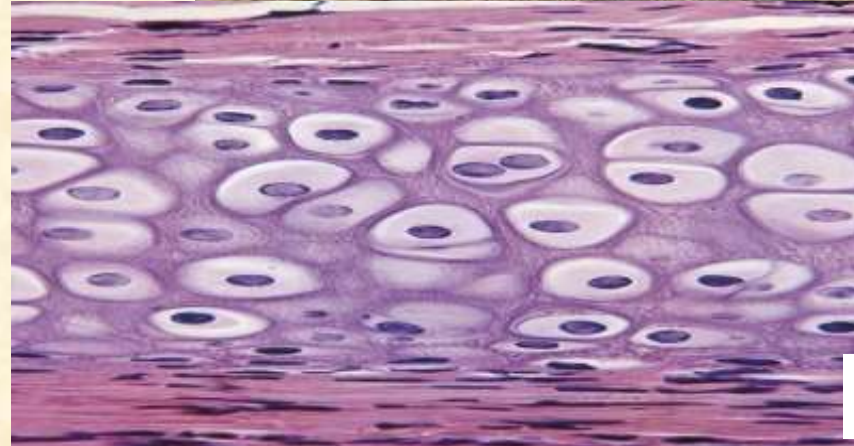
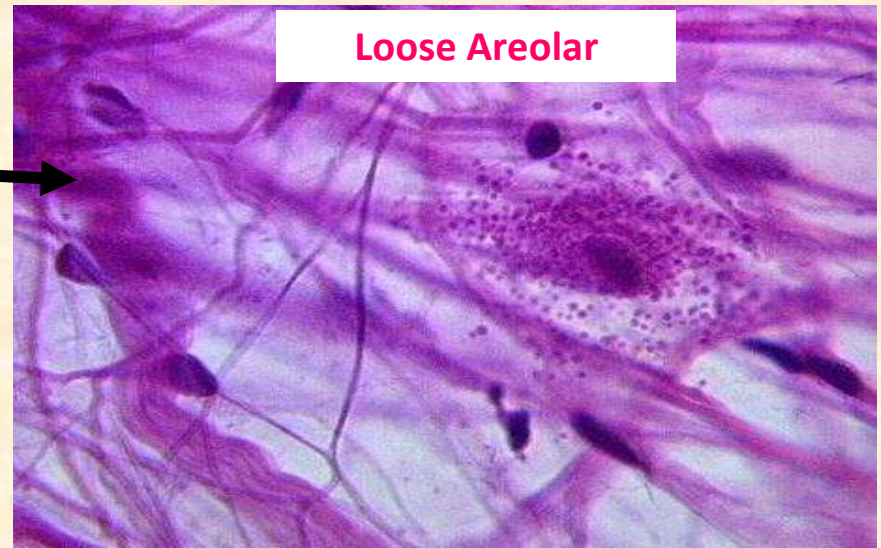
2. Dense Connective Tissue

- Fibers are predominant constituents.
- Much lower number of cells and less ground substance.
- Fibers appear closely packed in tissue sections.

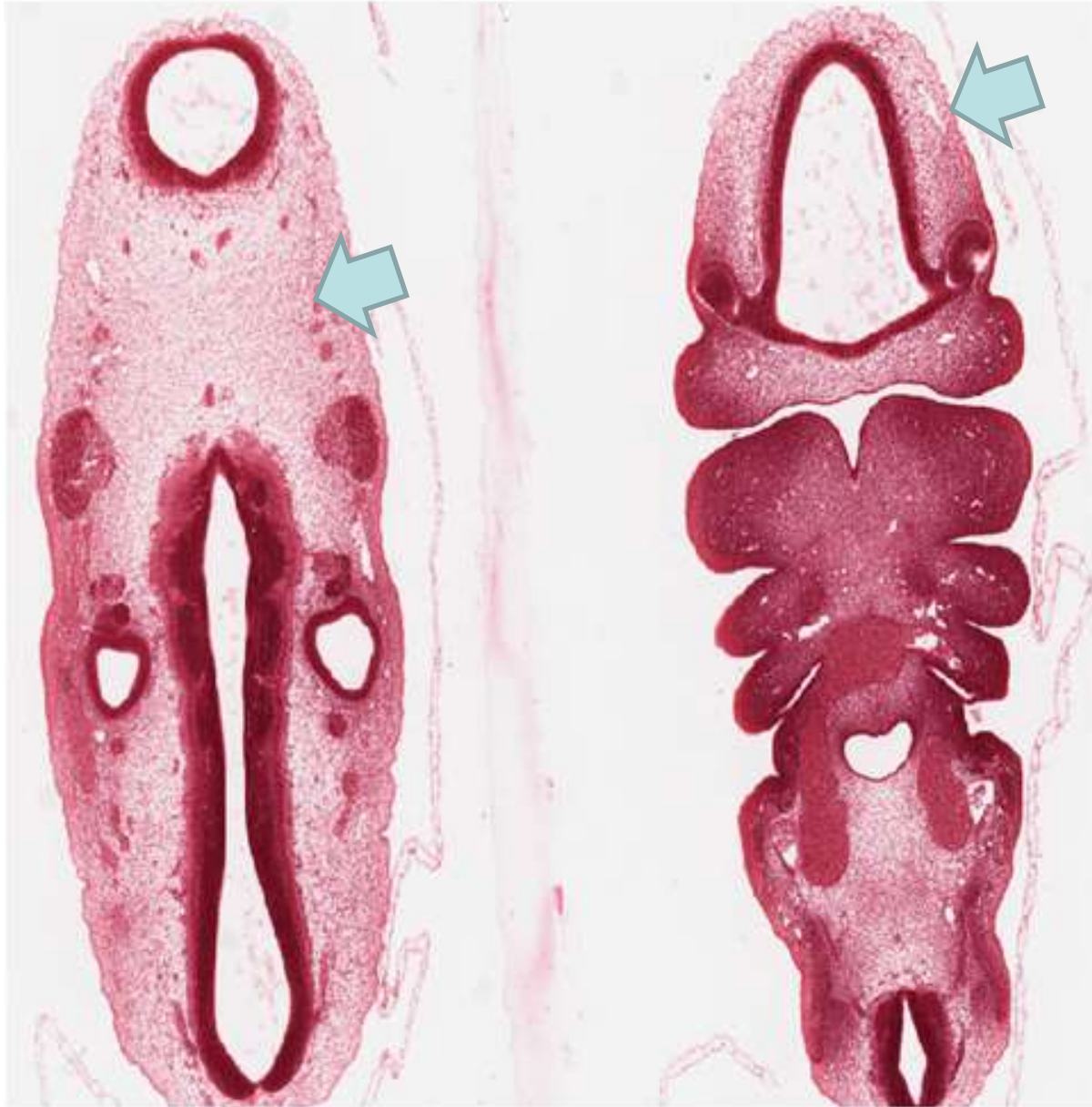


Categories of CT

- Fibrous connective tissue (a.k.a. connective tissue proper)
- Supportive/skeletal connective tissue
- Fluid connective tissue (Blood)



Mesenchyme, in embryonic body wall



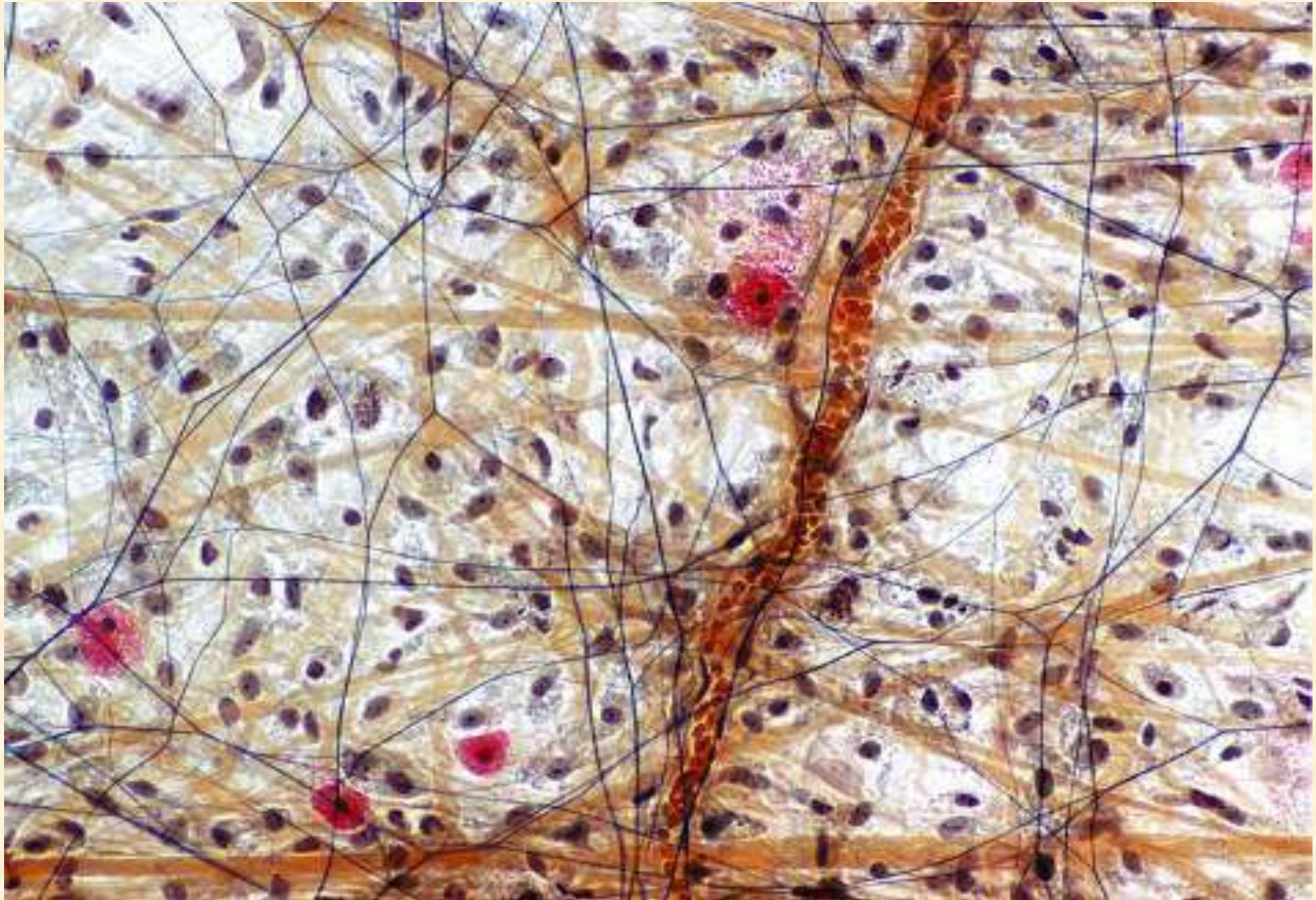
Loose Areolar CT

- The most numerous cells are fibroblasts and macrophages, but all other types of ordinary CT cells are also present.
- less amount of all CT fibers also appear in this tissue.
- With a moderate amount of ground substance, loose areolar connective tissue has a delicate consistency; it is flexible, well vascularized, and not resistant to stress.

Locations

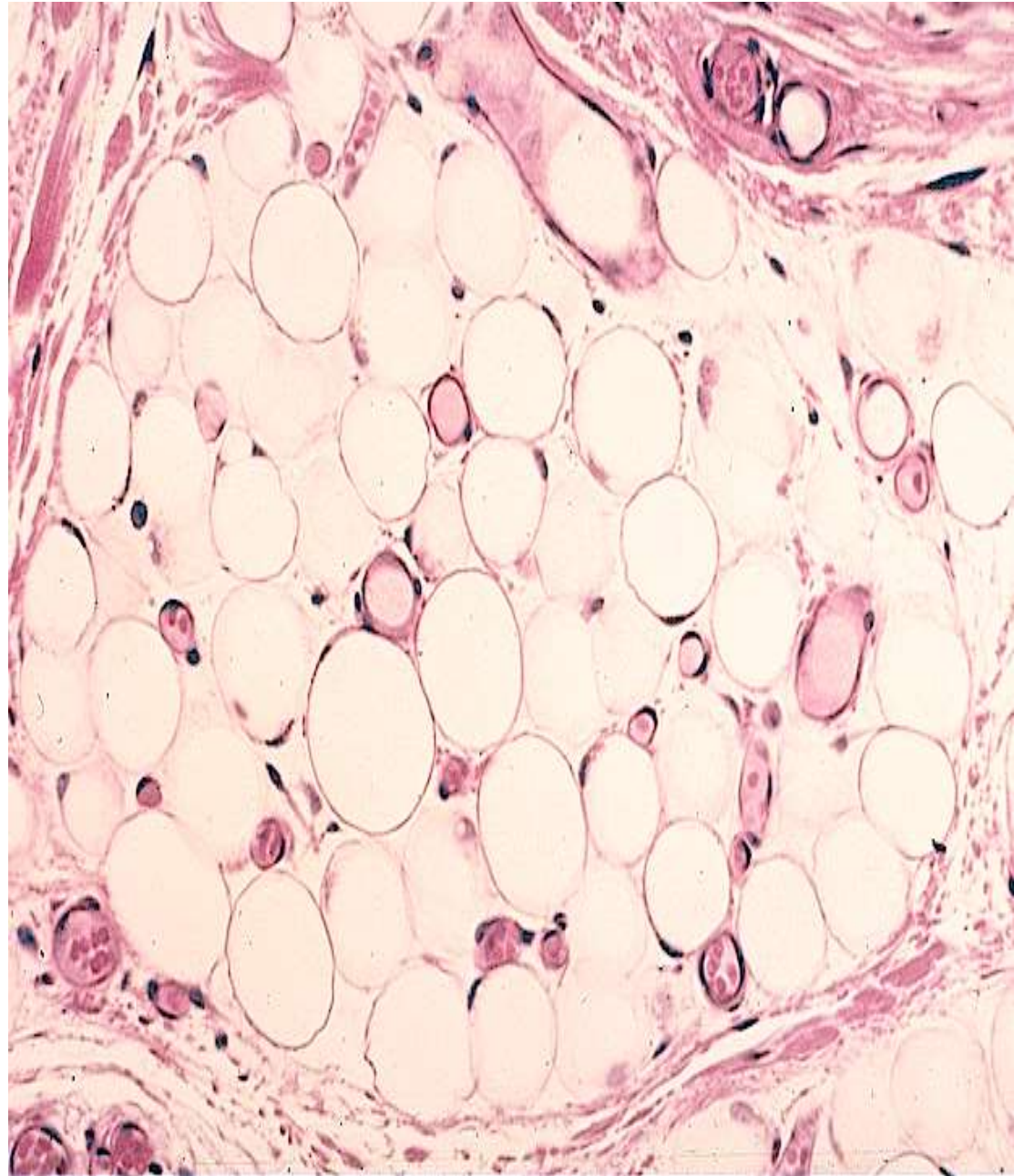
- in papillary dermis of skin & subcutaneous tissue,
- lamina propria & submucosa of GIT & respiratory tracts,
- around blood vessels & organs and serous membranes, around muscle fibers (endomysium) and nerve fibers (endoneurium).
- accumulates large amount of tissue fluid and leads to tissue swelling (Edema) in inflammatory conditions.

LOOSE AREOLAR CT



Functions of Adipose Tissue

- Energy storage
- Thermal insulation
- Shock absorption
- Protective cushioning for some organs
- Accounts 15–20% of the body weight in men of normal weight; in women of normal weight, 20–25% of body weight.



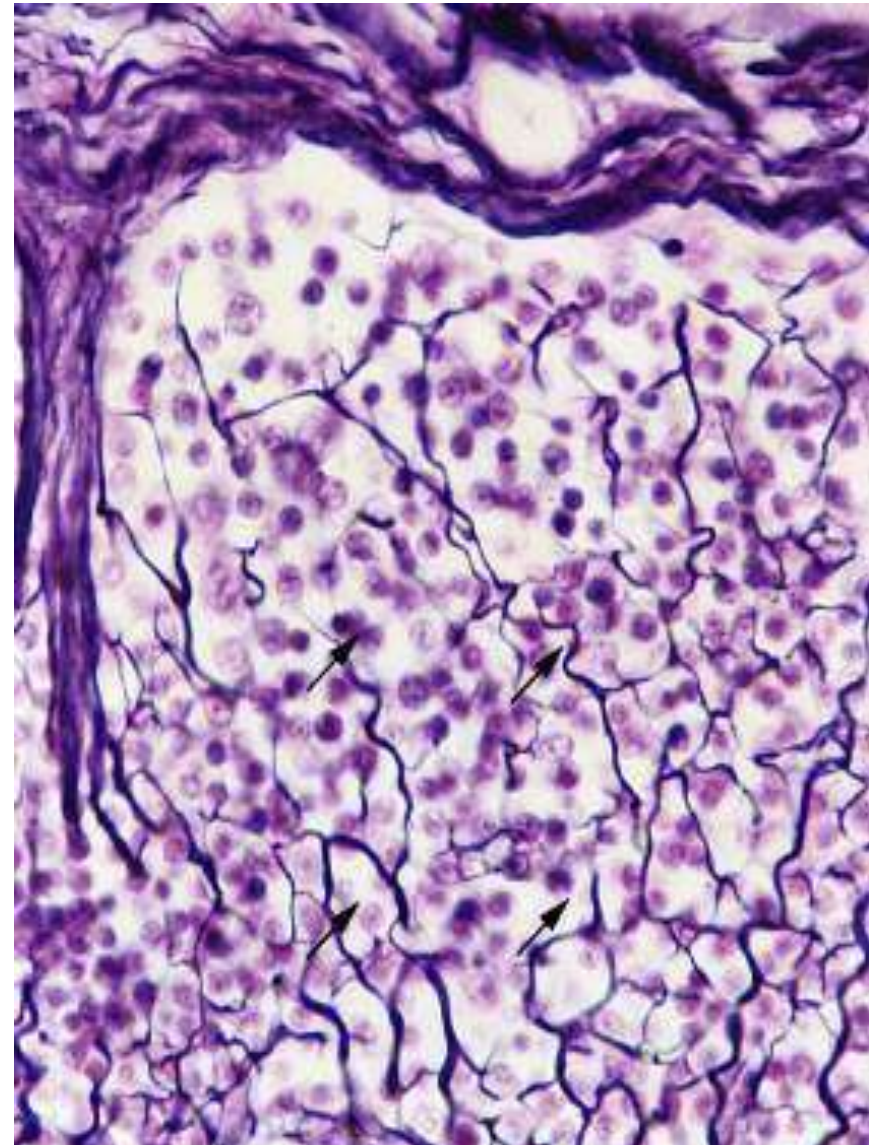
White and Brown Adipose Tissues

- The color of brown adipose tissue or **brown fat** is due to both the numerous mitochondria found in the adipocytes and the large number of blood capillaries in this tissue.
- Adipocytes of brown fat are therefore called **multilocular**
- The many small lipid droplets, abundant mitochondria, and rich vasculature all help mediate this tissue's principal function of **heat production**.
- unlike white adipose tissue, which is present throughout the body, brown adipose tissue has a much more limited distribution.



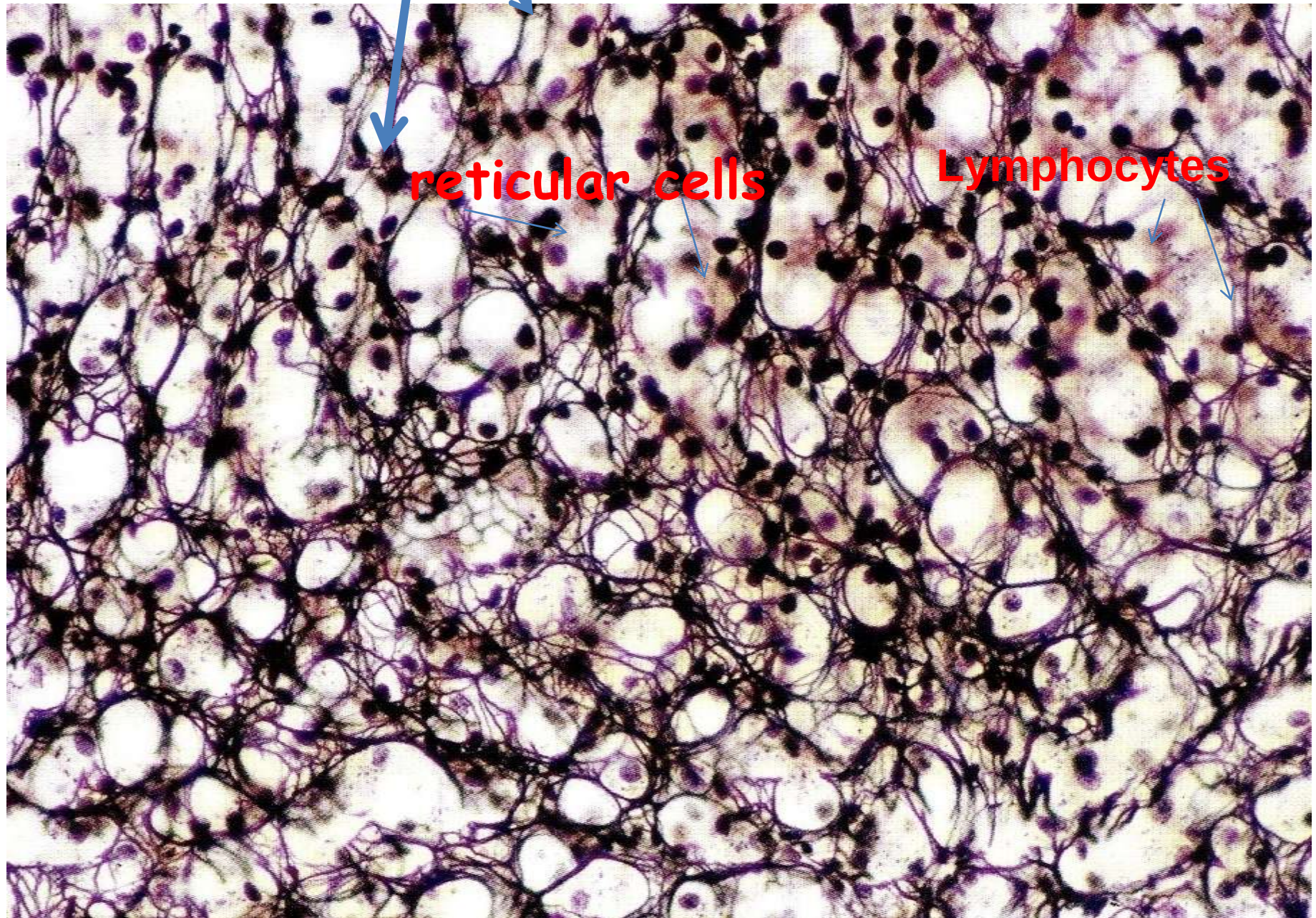
Reticular Ct

- composed mainly of **type III collagen fibers** in a form of network or mesh, fibroblasts called reticular cells, and lymphocytes.
- Forms supportive framework of liver sinusoids, red bone marrow, lymph nodes, spleen, smooth muscles, islet of Langerhans and adipose tissues.
- **Reticular cells** - synthesize reticular fibers in lymphatic tissues; cell body and processes completely surround reticular fibers



Lymph node (silver stain)

RETICULAR CT



2. DENSE CT

- Contains abundant amount of fibers gathered into large bundles with fewer cells & ground substance.

a. DENSE REGULAR CT

- Has tightly packed coarse collagen bundles arranged in parallel fashion to form cord-like cylinders (tendons, retinacula & ligaments) or sheets (aponeurosis) & capsule of some glands (spleen, suprarenal, etc), interosseous membranes, fibrous sheaths around cartilage (perichondrium) and sheaths around bone (fibrous periosteum).
- Fibroblasts and fibrocytes are arranged in rows between the collagen fibers so they appear elongated with their dash-like elongated nuclei.
- It can be collagenous type (e.g. tendon, aponeurosis, retinacula, white ligaments) or elastic type (e.g. ligamentum nuchae, ligamentum flavum)
- Dense collagenous CT type is also known as **white fibrous CT**.

Dense Regular Collagenous C.T

Tendon, teased



Dense Regular Connective Tissue

- The collagen bundles are arranged according to a definite pattern, with the linear orientation of fibroblasts in response to prolonged stresses exerted in the same direction.
- This arrangement offers great resistance to traction forces.



NOTE the waviness of the fibers. What function could this structural aspect provide?

b. DENSE IRREGULAR CT

- Contains abundant amounts of fibers with less cells and ground substances.
- Has tightly packed coarse collagen bundles arranged randomly in all directions, with few elastic fibers.
- Fibroblasts and other fewer cells are arranged between the collagen fibers.

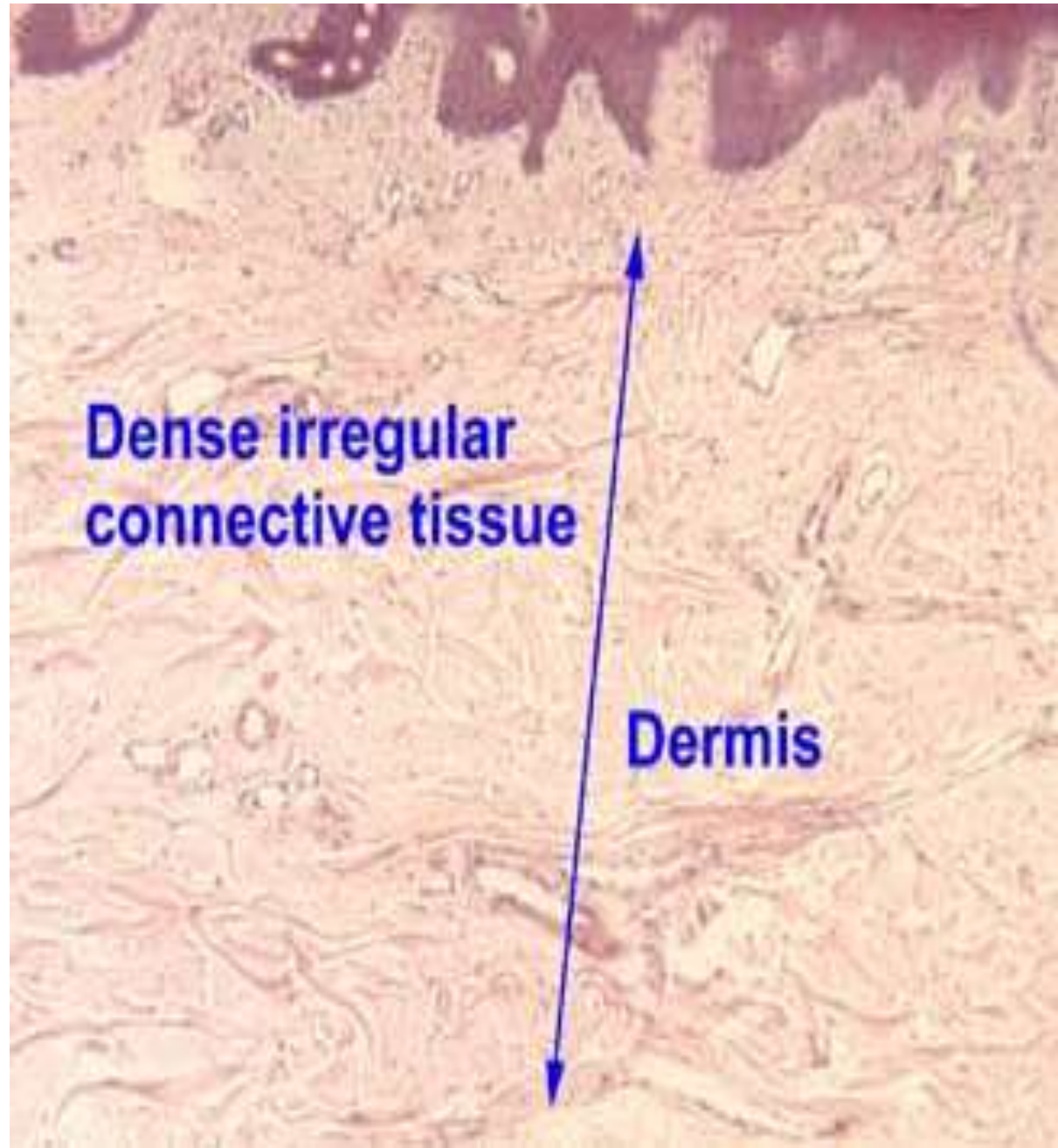
Location

- reticular dermis (the deeper 4/5th of the dermis of skin),
- capsule of organs, trabecula of glands & fibrous joints
- **Corpus Albicans (L., white body):** a scar of white dense fibrous connective tissue in ovary
- Sheaths around skeletal muscles (epimysium), nerves (epineurium) and CNS (dura & arachnoid maters).

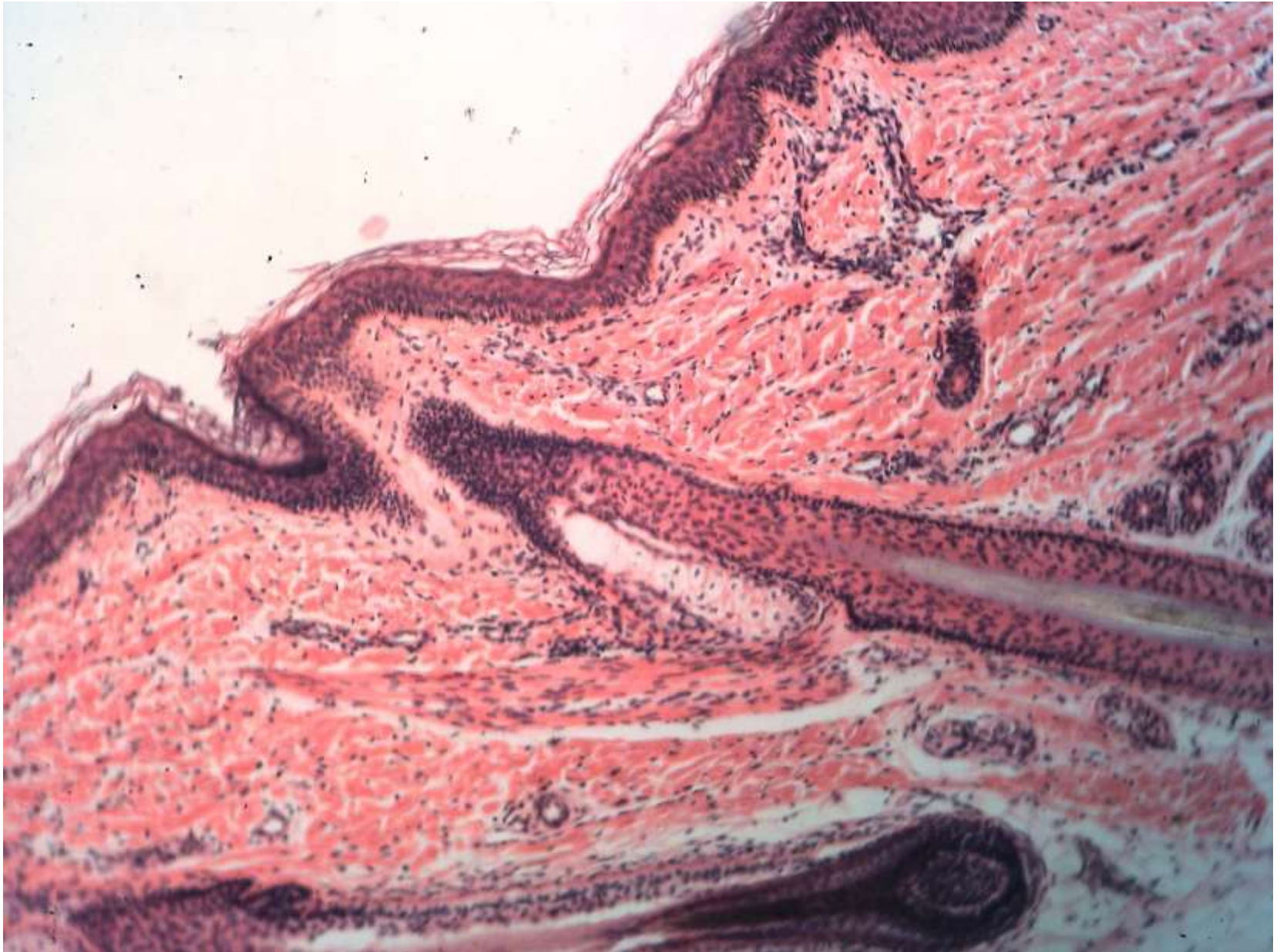
Locations:

- Deeper portion of dermis of skin.
- Capsules around some organs such as testes, lymph nodes, ganglia, sensory receptors, and trabeculae of some organs like spleen lymph node and glands.
- Stroma of some glands like inactive mammary gland

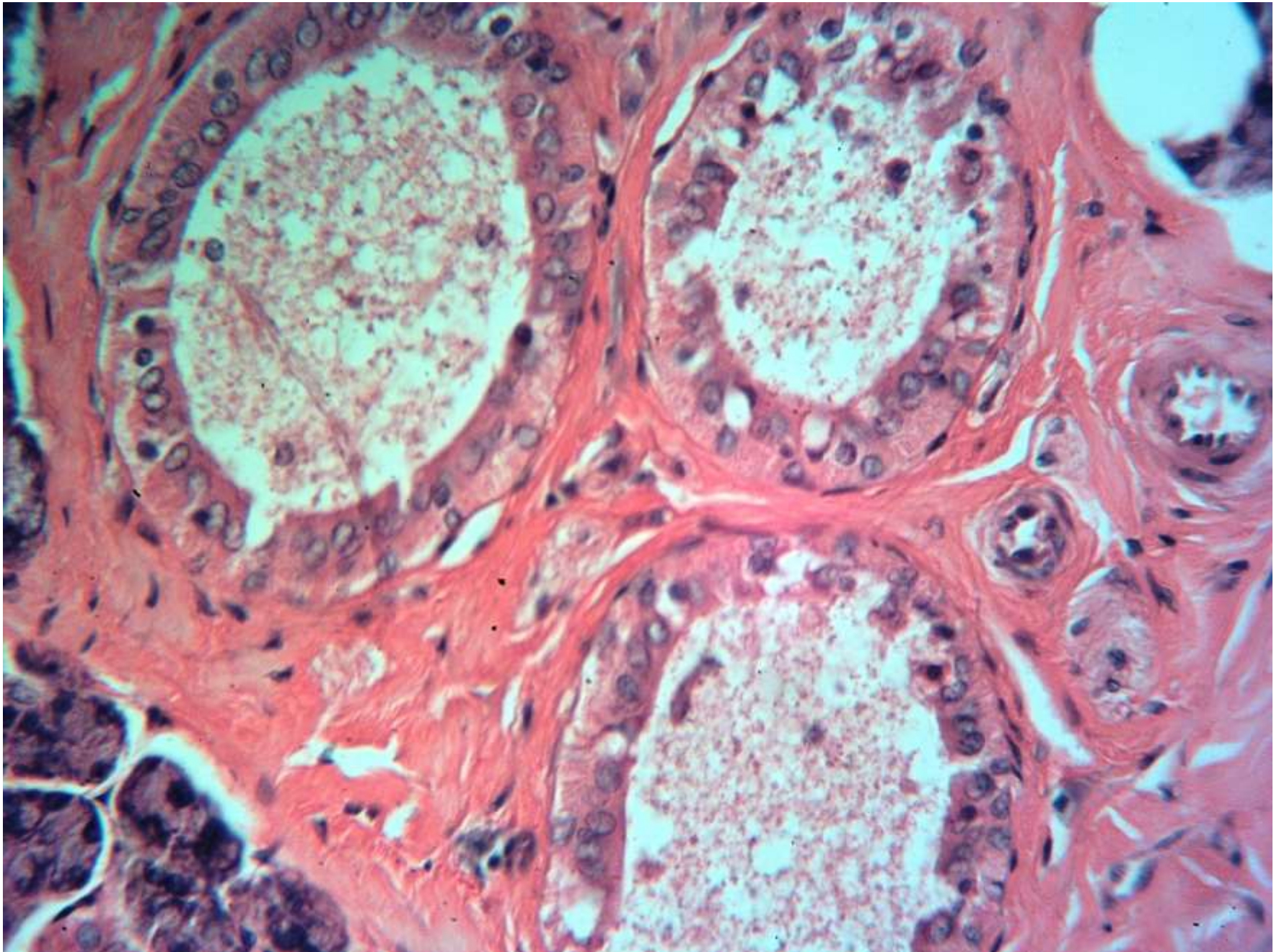
Dense Irregular CT



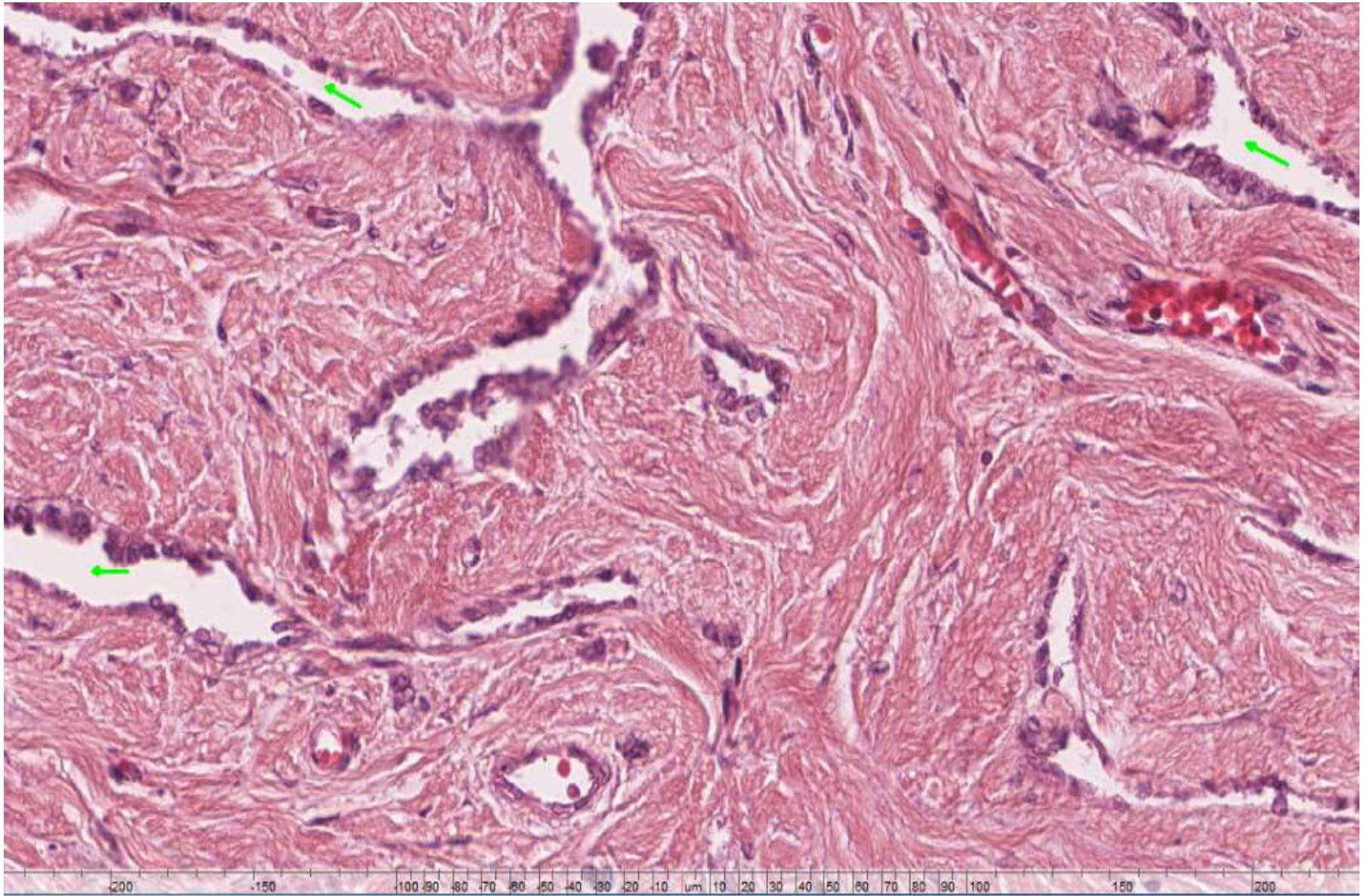
Dense Irregular CT of the lower dermis



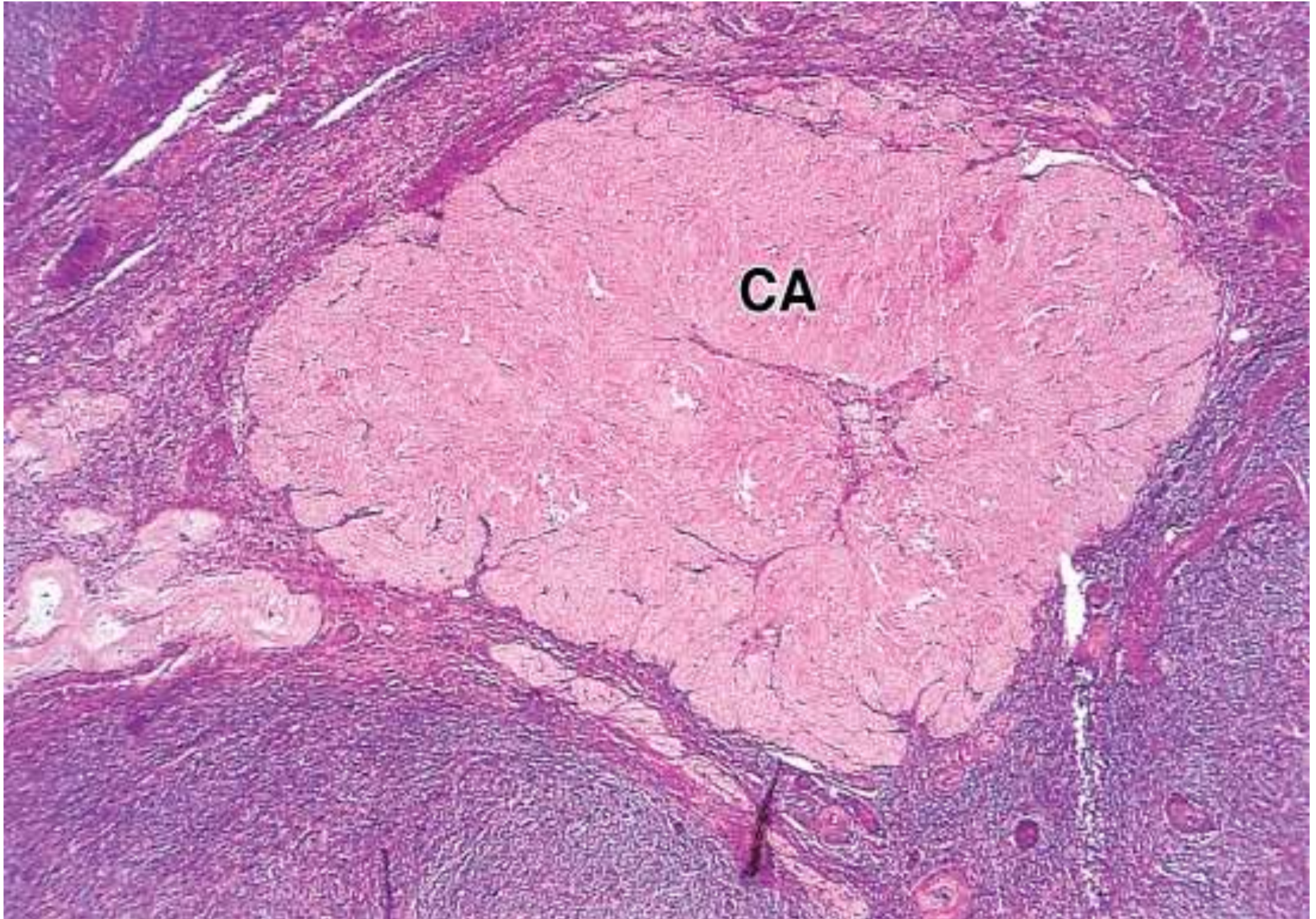
Dense Irregular CT trabecula of submandibular gland



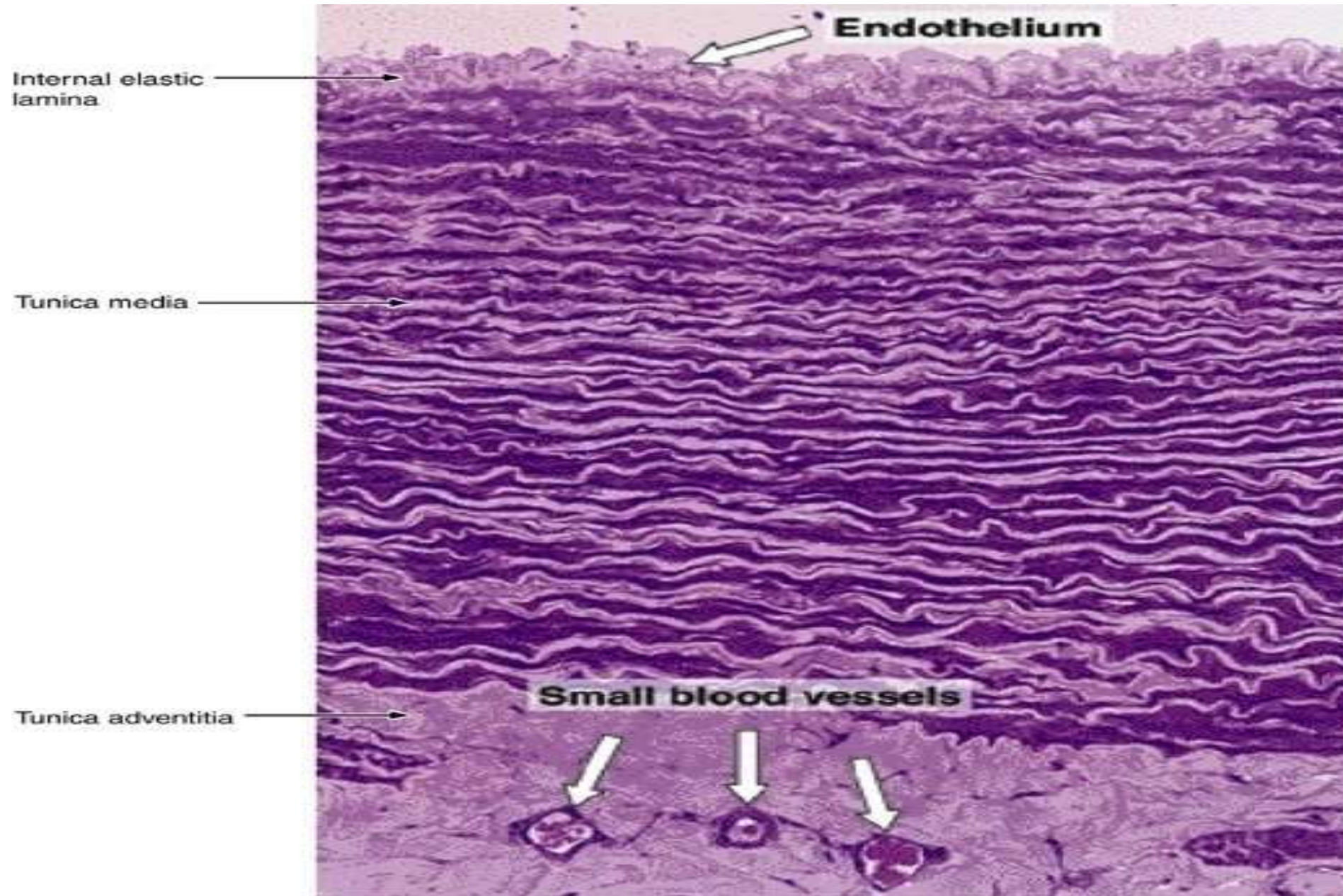
Dense Irregular CT of mediastinum testis



Corpus Albican (CA)



Elastic C T in the wall of large artery (Aorta)

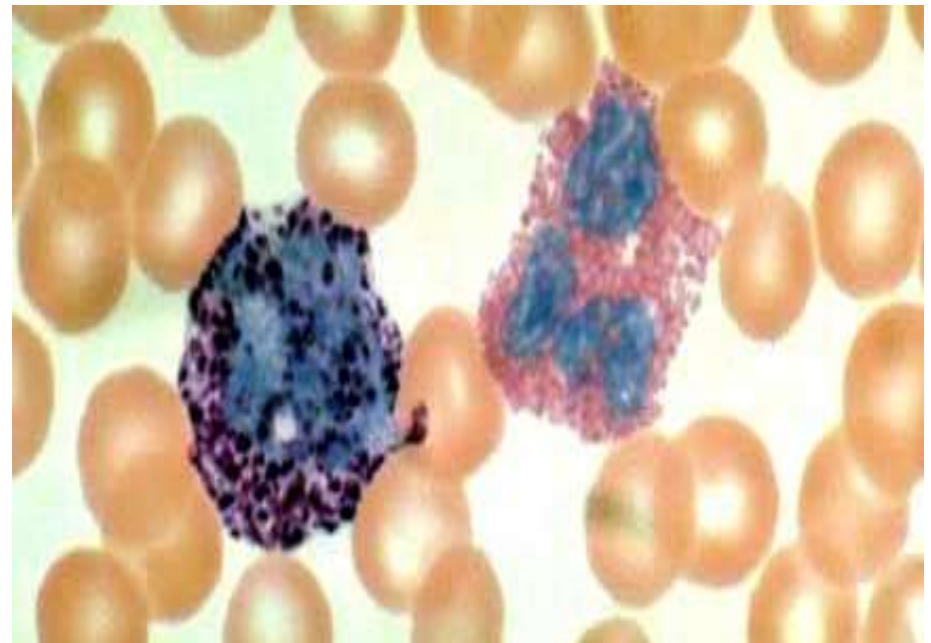
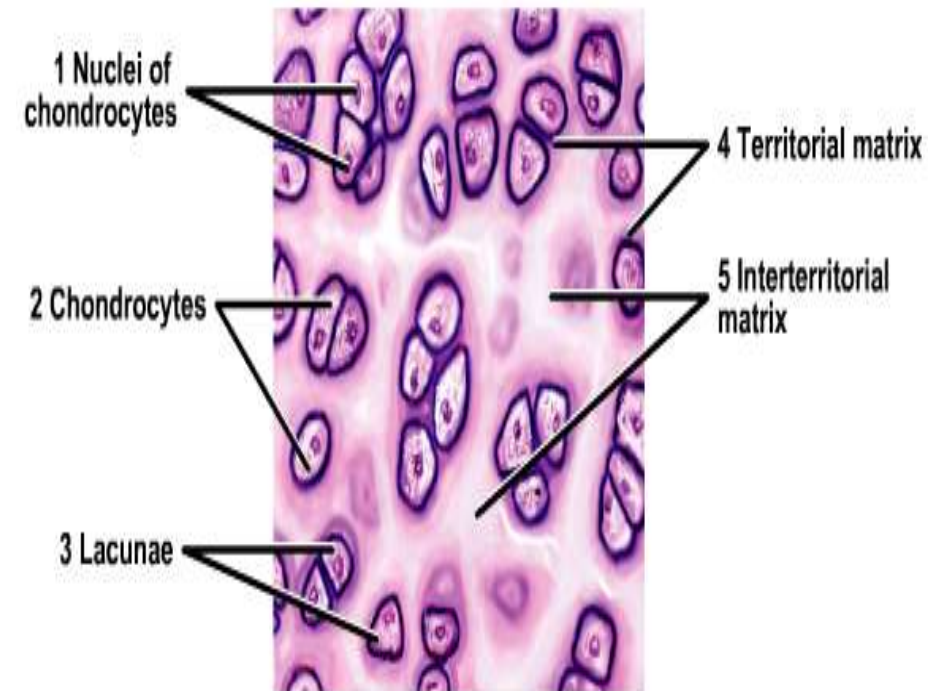


CONNECTIVE TISSUES WITH SPECIAL PROPERTIES

1. BONE TISSUE

2. BLOOD TISSUE

3. CARTILAGE



CARTILAGE

- belongs to the skeletal (supportive) tissues which provide flexible support to some organs.

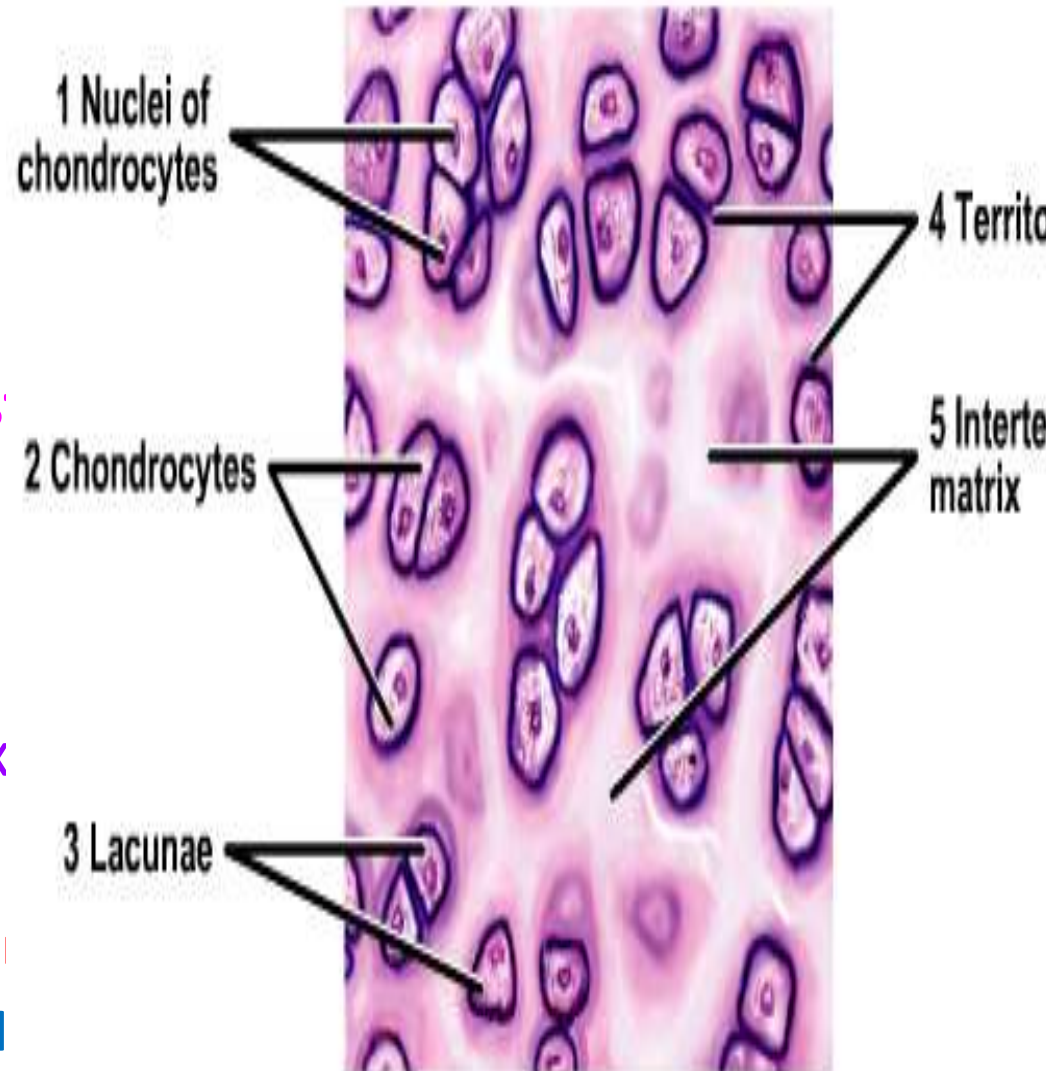
Cartilage consists of:-

1- Cartilage cells: chondroblasts and chondrocytes

- **chondrocytes** are located in **lacunae** surrounded by an intercellular cartilage matrix

2. Extracellular matrix:

- Collagenous and elastic fibers**
- Ground substance: enriched with glycosaminoglycans and proteoglycan aggregates.**



Characteristics of Cartilage

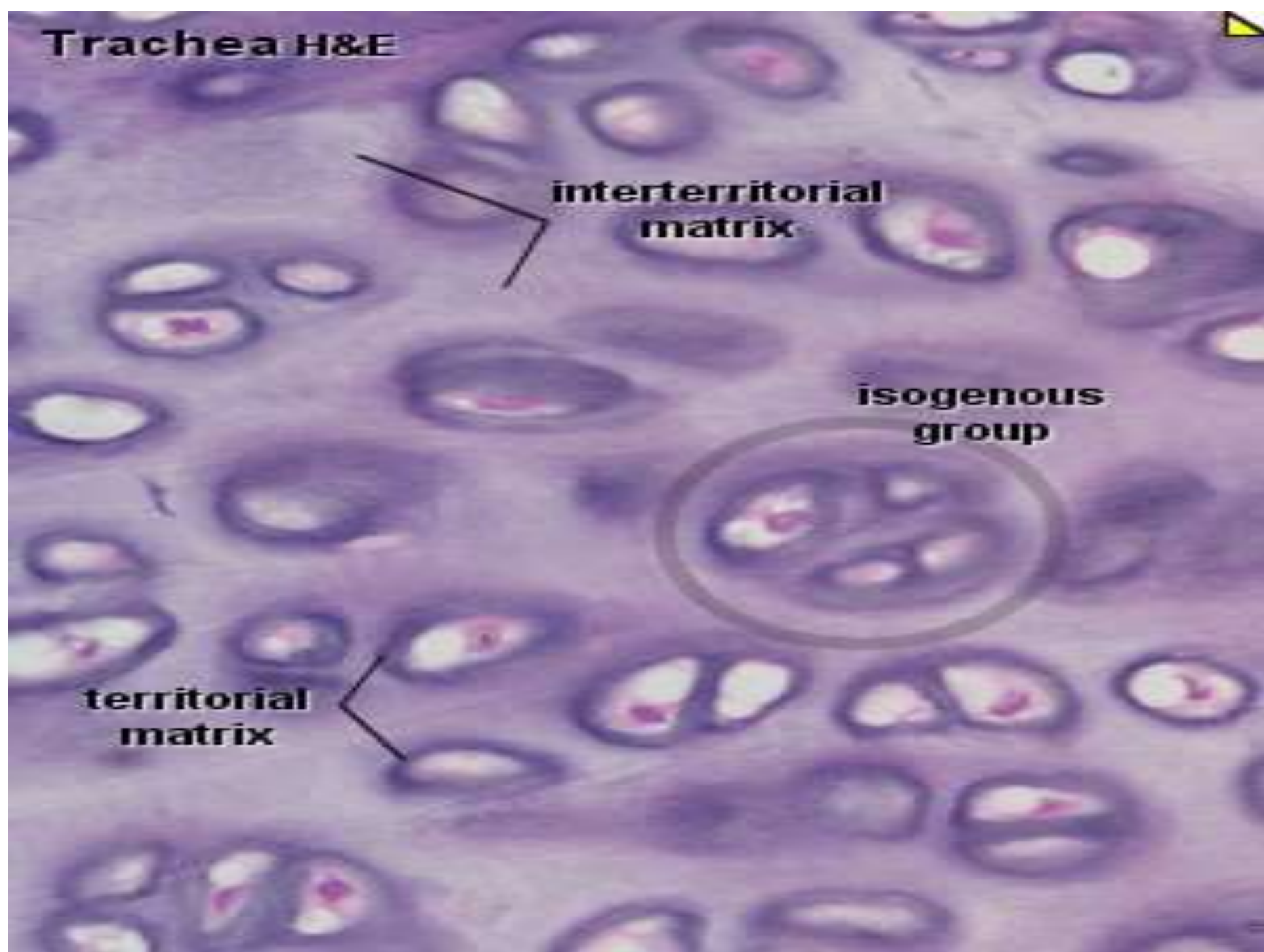
1. Cartilage is avascular (has no blood supply) so, healing is poor or slow after injury.
2. It has no lymphatic vessels or nerves either.
3. **Chondroblasts** and **chondrocytes** synthesize and secrete the cartilage matrix.
4. **Chondrocytes** are located in matrix cavities called **lacunae**.
5. Chondrocytes derived by mitotic division of one parent cell exist as clusters of 4 to 8 cells known as **isogenous groups (cell nests)**.
6. Cytoplasm of chondrocytes is basophilic, rich in glycogen, lipids and phosphatase enzymes.
7. Cartilage is usually covered or surrounded by a capsule-like sheath of dense CT known as **Perichondrium**, formed of two layers: outer fibrous layer & inner chondrogenic layer.

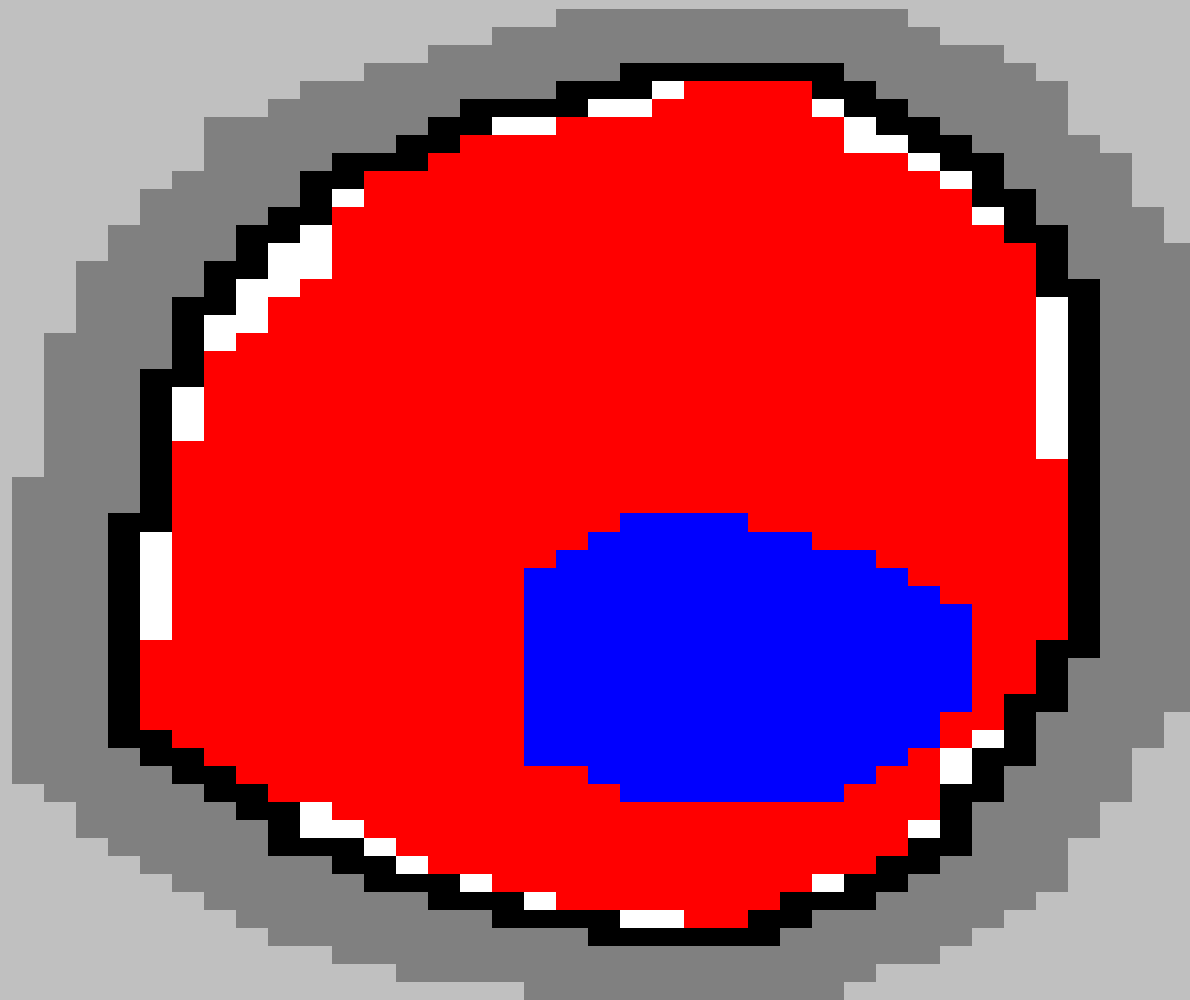
Trachea H&E

**interterritorial
matrix**

**isogenous
group**

**territorial
matrix**





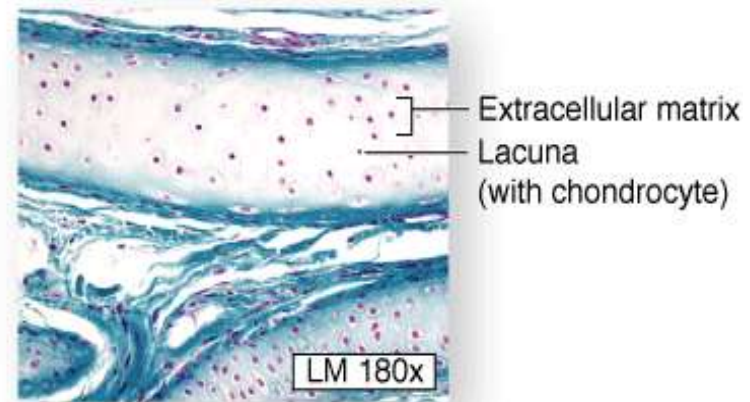
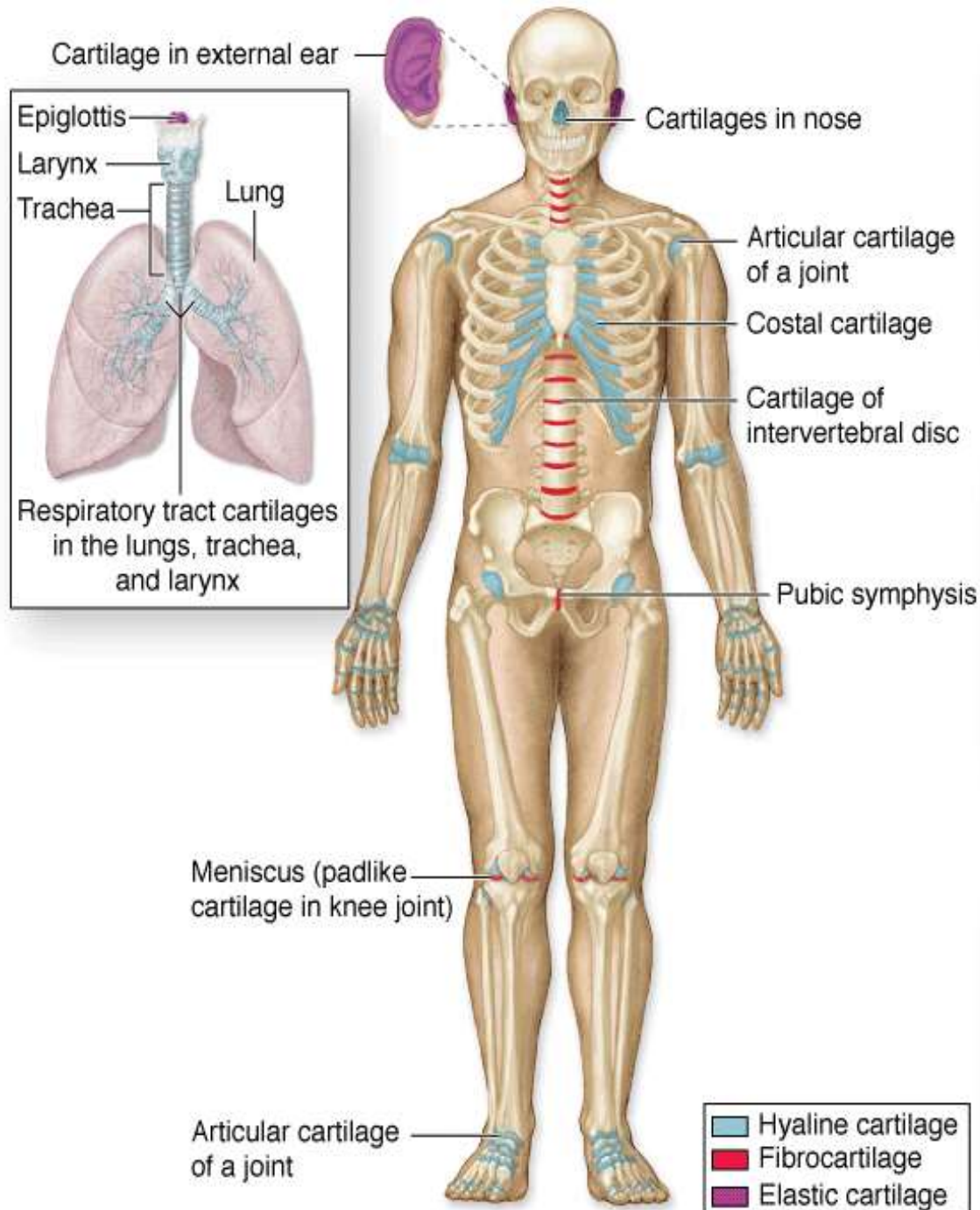
1st cycle

- Variations in the composition of cartilage matrix components produce three types of cartilages adapted to local biomechanical needs:

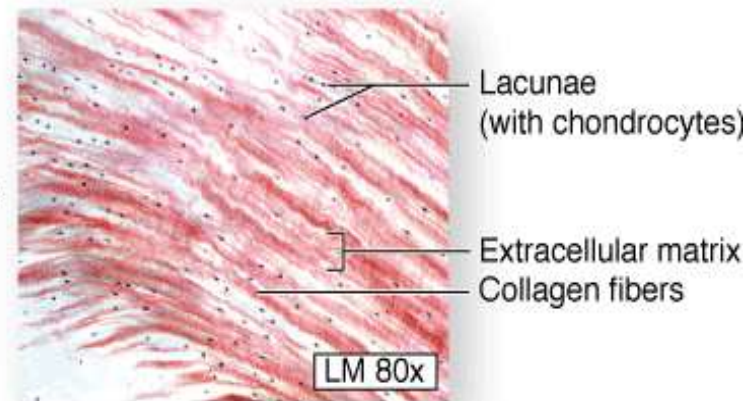
- 1. Hyaline cartilage**

- 2. Elastic cartilage**

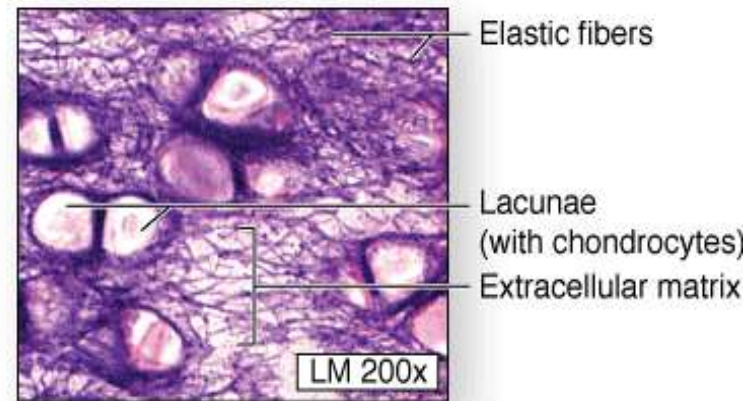
- 3. Fibrocartilage**



b Hyaline cartilage



c Fibrocartilage



d Elastic cartilage

- **Muscular tissue**

The cells of muscle tissue are long narrow called **muscle fibers**

The cell membrane of muscle fiber is called **sarcolemma**, the cytoplasm is called **sarcoplasm**, the smooth endoplasmic reticulum is called **sarcoplasmic reticulum**, the mitochondria is called **sarcosome**

Muscle fibers are either **striated** or **non-striated** according to whether or not they have transverse striations

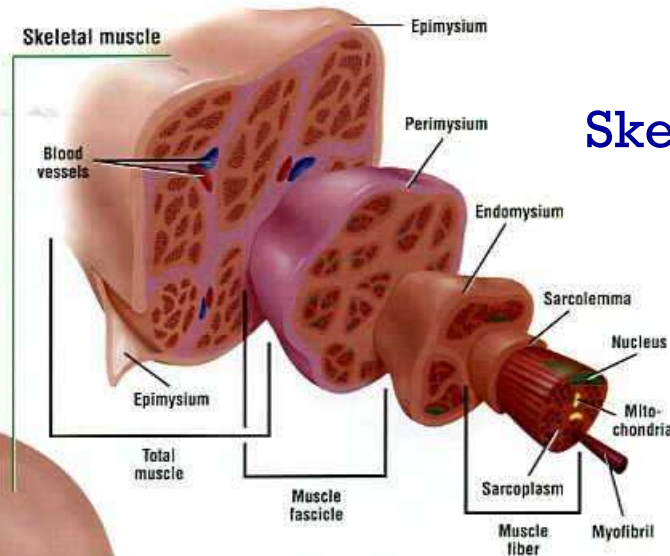
Muscle fibers are either **voluntary** or **involuntary**

Types of Muscular Tissue:

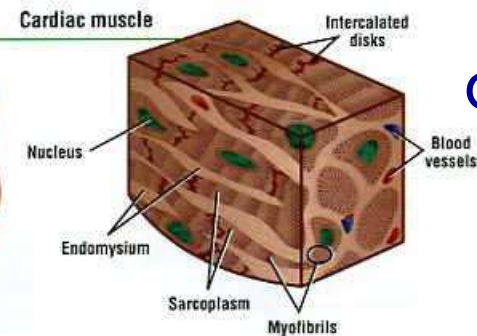
- Skeletal muscle
- Cardiac muscle
- Smooth muscle



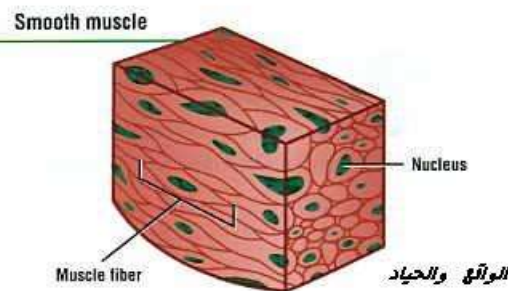
Skeletal muscle

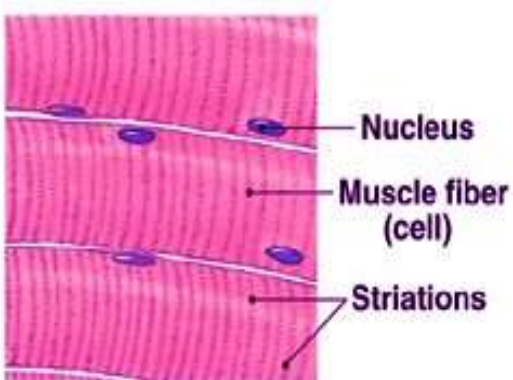
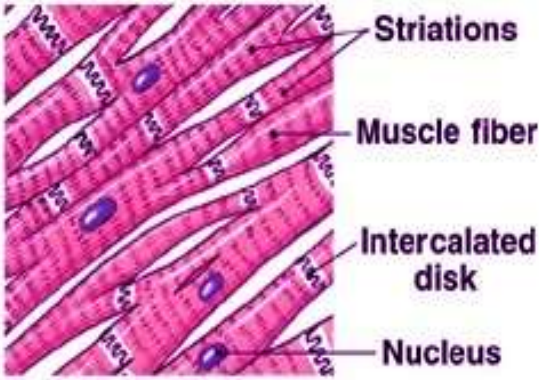
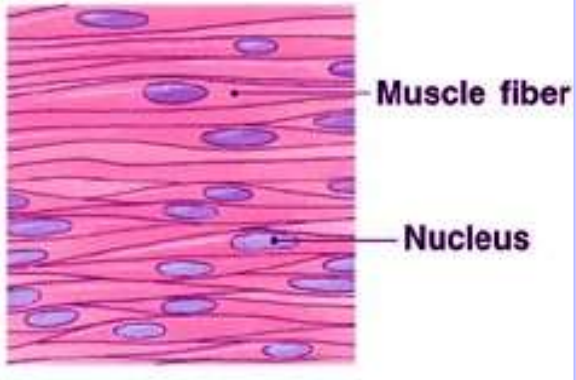


Cardiac muscle



Smooth muscle



<i>Skeletal muscle</i>	<i>Cardiac muscle</i>	<i>Smooth muscle</i>
Attached to the skeleton	Found in the myocardium of the heart	Found in the blood vessels & viscera
Striated	Striated	Non-striated
Voluntary	Involuntary	Involuntary
 This diagram illustrates skeletal muscle tissue. It shows several long, cylindrical muscle fibers arranged in parallel. Each fiber contains multiple, small, dark-stained nuclei located along the periphery. The fibers exhibit prominent transverse striations, which are alternating light and dark bands. Labels with leader lines point to a 'Nucleus', a 'Muscle fiber (cell)', and 'Striations'.	 This diagram illustrates cardiac muscle tissue. It shows branched muscle fibers arranged in a somewhat regular pattern. The fibers are striated and contain a single, centrally located nucleus. Intercalated disks, which are specialized junctions between adjacent cells, are visible as dark, wavy lines. Labels with leader lines point to 'Striations', 'Muscle fiber', 'Intercalated disk', and 'Nucleus'.	 This diagram illustrates smooth muscle tissue. It shows spindle-shaped muscle fibers arranged in a somewhat regular pattern. The fibers are non-striated and contain a single, centrally located nucleus. Labels with leader lines point to a 'Muscle fiber' and a 'Nucleus'.

- **Nervous tissue**

The nervous tissue is the basic tissue from which the nervous system is made

The nervous system includes:

- **Central nervous system:** the brain and spinal cord
- **Peripheral nervous system:** peripheral nerves (spinal & cranial nerves) and nerve ganglia (spinal ganglion & autonomic ganglion)

Histologically, the nervous tissue is formed of **neurons** and **neuroglia**

The Neuron is the structural and functional unit of the nervous system.

Neuroglia is the interstitial tissue of the nervous system. It consists of large number of branching cells which bind and connect the neurons together and with the blood vessels

Central and peripheral nervous system

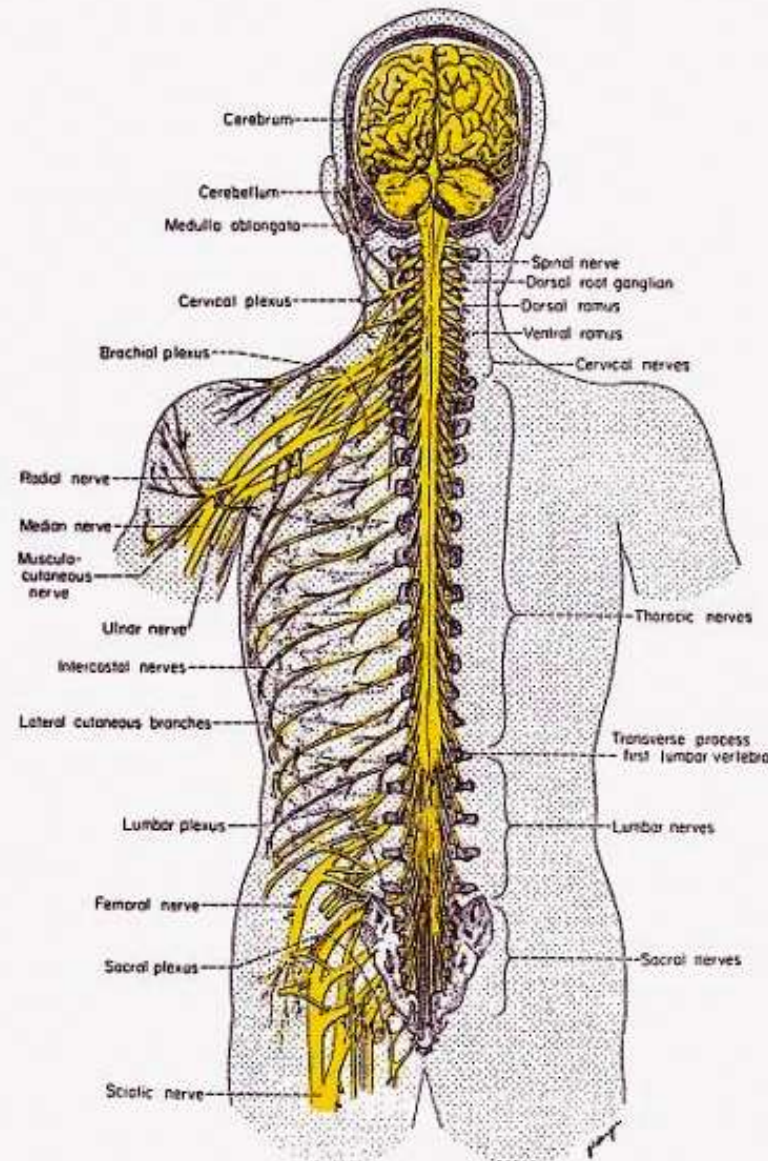


Fig. 2. The human central nervous system, exposed by dissection from the dorsal aspect. Shows the brain, spinal cord and the proximal parts of the spinal nerves.

The Neuron

The neuron is the structural and functional unit of the nervous system

The neuron consists of:

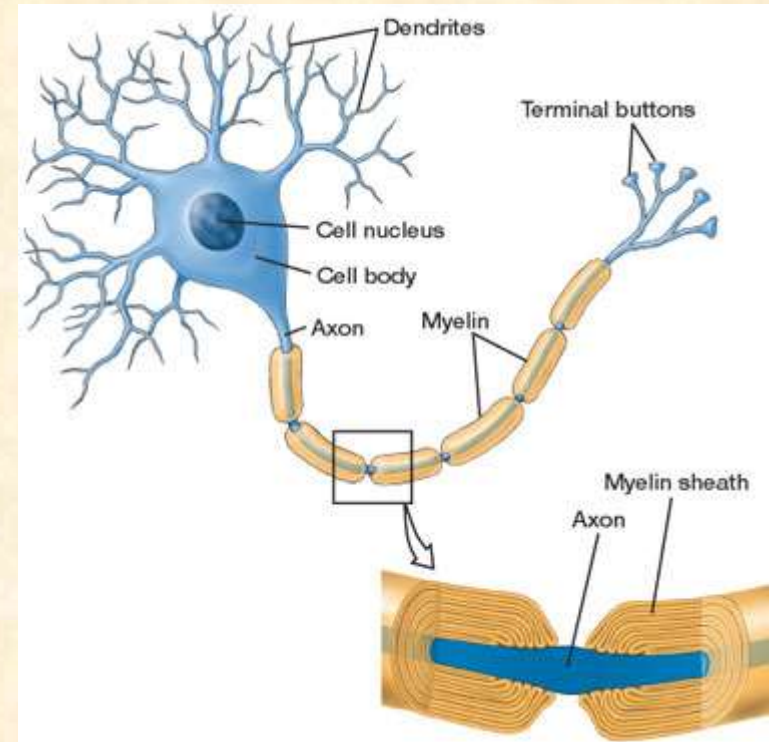
- Cell body
- Processes (Axon & Dendrites)

The Neuroglia

They are classified into neuroglia proper & other types of neuroglia

Neuroglia proper:

1. Astrocytes
2. Microglia
3. Oligodendroglia



GOOD LUCK